Is a Description Worth a Thousand Demonstrations? Measuring the Effective Sample Size of Task Descriptions in In-Context Learning

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Abstract

Prompts for in-context learning usually pair demonstrations with a task description, yet theories of in-context learning mostly model the demonstrations alone. We ask how many demonstrations a description is worth: its *effective sample size* (ESS), the number of demonstrations that lower a learner's error as much as the description does. For the Bayes-optimal predictor in in-context linear regression, the ESS depends on the description's *specificity* (how much uncertainty about the task it removes) and its *reliability* (the probability that it is relevant). We prove that a reliable description's ESS grows linearly with the task dimension when the description is loose and with its precision when it is specific, and that a description of only part of the task is worth about the number of directions it covers until demonstrations resolve the rest. Without demonstrations, a description that may be irrelevant is worth a bounded number of demonstrations however specific it is; with them, trusting it fully can be worse than ignoring it. Meta-trained Transformers come close to the Bayes-optimal ESS and take their trust from training. Pretrained language models answering without reasoning adjust to neither the stated specificity nor the stated reliability as the Bayes-optimal predictor does: with the stated value held fixed, they respond to its stated precision far less than it does, and a description worth 36 demonstrations to it is worth 1 to 8 to them. Models that reason first follow the stated specificity, and two come close to the Bayes-optimal ESS.

1. Introduction

Prompts often tell a model what the task is before, or instead of, showing it. A few-shot prompt for in-context learning (ICL) pairs a natural-language task description with demonstrations (Brown et al., 2020), and deployed models also receive descriptions as system prompts, task instructions, and the instruction files that configure coding agents (Chatlatanagulchai et al., 2025). Yet we know little about how a model uses such a description or how effective it is in different forms, and the evidence is mixed. A prompt can be worth hundreds of data points in prompt-based fine-tuning (Le Scao & Rush, 2021), and a written summary can stand in for many demonstrations (Honda et al., 2025). Models, however, learn as fast from irrelevant or misleading instructions as from instructive ones (Webson & Pavlick, 2022), let in-context evidence outweigh explicit instructions (Gupta et al., 2025), and coding agents given repository instruction files do not generally solve more tasks (Gloaguen et al., 2026).

Theories of ICL, in contrast, mostly model the prompt as a sequence of demonstrations (Xie et al., 2022; Garg et al., 2022; Akyürek et al., 2023); those that add a description show that a learner can use it to predict better (Huang & Ge, 2025), but neither put a value on it nor say what it is worth when it may be wrong. We measure the value of a task description by the number of demonstrations it can replace, and ask what determines that number.

We take the Bayesian view of ICL (Xie et al., 2022): the prompt is evidence about a latent task, a demonstration is one noisy observation of it, and a description is a statement about which task generated the demonstrations, so it sharpens the prior; we set aside instruction following. Bayesian statistics measures the information in a prior by its effective sample size (ESS), the number of observations that would carry the same information (Morita et al., 2008). We define the ESS of a description in the same way (Figure 1)

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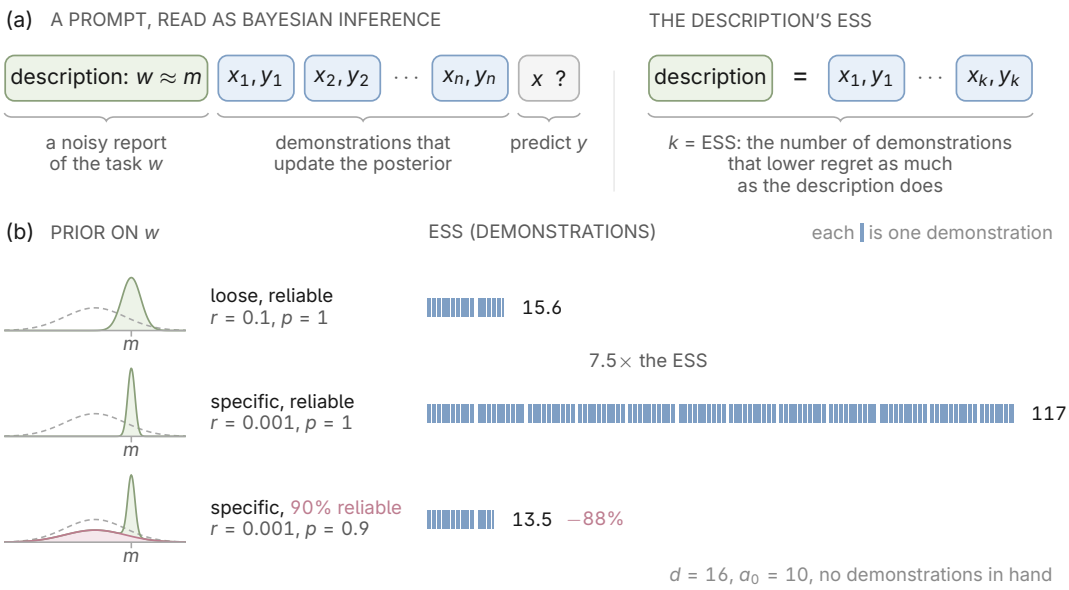


Figure 1. **A description is worth a number of demonstrations, set by its specificity and its reliability.** (a) A prompt read as Bayesian inference: a description states that the task w is near a value m, and the demonstrations update the posterior. The ESS of the description is the number of demonstrations it replaces. (b) The prior on w without the description (dashed) and given three descriptions (solid; sketched, not to scale), and the ESS of each with no demonstrations in hand, at task dimension $d = 16$ and prior signal-to-noise ratio $a_0 = 10$; r is the fraction of the prior variance a relevant description leaves, p the probability that it is relevant, and pink marks the component for an irrelevant description. Specificity raises the ESS (§4); a one-in-ten chance of being irrelevant removes most of that gain (§5).

and compute it in noisy in-context linear regression with a Gaussian task prior (Garg et al., 2022; Akyürek et al., 2023; Raventós et al., 2023), where Bayes-optimal prediction has a closed form, and then measure it for meta-trained Transformers and for pretrained language models given a stated prior.

Following the Bayesian literature on informative priors, we characterize a description by two properties. The first is its *specificity*, the precision of the prior it gives (Morita et al., 2008; Neuenschwander et al., 2020). "Write this report like an accountant" only loosely indicates the task; a description that fixes the report's sections, figures, and accounting standard all but states it. The second is its *reliability*, the probability that it is relevant, which sets the weight of that prior in a robust mixture that guards against conflict with the data (Schmidli et al., 2014; Evans & Moshonov, 2006): instructions written for one report can be pasted into a prompt for another, or simply be wrong. Our contributions are:

- **A common unit.** Following data-dependent ESS for priors (Wiesenfarth & Calderazzo, 2020; Reimherr et al., 2021), we define the ESS of a task description, the number of demonstrations that lower a learner's error as much as the description does (Definition 1). It applies to any learner, and with squared error it needs only the learner's answers.
- **Theory.** For the Bayes-optimal predictor, a reliable description's ESS grows linearly in the task dimension when the description is loose and in its precision when it is specific (Proposition 1); the dimension term, absent from the information-based ESS of a prior, fades as demonstrations accumulate. A description of k of the d directions is worth about k without demonstrations, whatever its precision, and $(1 - r)ck/d$ with many (Proposition 2). With the law of total variance we show that, with no demonstrations in hand, a description that may be irrelevant is worth a bounded number however specific it is, and that once demonstrations are in hand, trusting it fully can be worse than ignoring it (Proposition 3, Corollary 4, Appendix A.4).
- **Experiments.** Meta-trained Transformers come close to the Bayes-optimal ESS and take their trust from training (§6). Pretrained language models answering without reasoning adjust to neither a description's stated specificity nor its stated reliability as the Bayes-optimal predictor does, even when only the stated specificity changes: a description worth 36 demonstrations to it is worth a median of 1 to 8 to them. Models that reason first follow the stated specificity, and Claude Haiku 5.5 and GPT-5.6-Luna come close to the Bayes-optimal ESS (§7, Appendix C).

2. Related Work

ICL as Bayesian inference. Xie et al. (2022) cast ICL as implicit Bayesian inference. Transformers trained on samples from a prior approximate its posterior predictive

(Müller et al., 2022), and meta-trained Transformers have been shown to implement Bayes-optimal estimators such as ridge regression under a Gaussian task prior, given enough task diversity (Akyürek et al., 2023; Raventós et al., 2023; Panwar et al., 2024).

Task descriptions in theories of ICL. Some theories of ICL include task descriptions, as tokens that indicate a hypothesis class (Lin et al., 2025) or carry the input mean (Huang & Ge, 2025), or as instructions in a generalization bound (Tong et al., 2026). They generally do not value a description in demonstrations or model the chance that it is irrelevant. Closest to our setting, Zhu et al. (2026) show that a Transformer given a prefix of related datasets performs amortized hierarchical Bayesian prediction; we express the value of such conditioning information in demonstrations and allow it to be irrelevant.

Effective sample size of a prior. Our ESS follows Bayesian methods for the sample size of a prior (Morita et al., 2008; Neuenschwander et al., 2020; Reimherr et al., 2021). As in Reimherr et al. (2021) and the effective current sample size of Wiesenfarth & Calderazzo (2020), it depends on the number of observations in hand, and as in Reznik (2026), who sets a power prior's discount on possibly mismatched historical data, it measures information by predictive loss. With many observations in hand it returns to the ESS of Morita et al. (2008) (Corollary 4(ii)); with few, matching errors rather than information adds a dimension term that theirs lacks (§4). To model a description that may be irrelevant we use a robust mixture prior, which clinical trials use to discount historical data that may conflict with the current trial (Schmidli et al., 2014).

Descriptions and priors in pretrained language models. Cook et al. (2026) find that one instruction can stand in for up to 100 execution examples in fine-tuning data, and Wenliang et al. (2026) that an optimized prompt can be shorter than demonstrations of equal effect. Tests of pretrained models against a known optimum find that in-context evidence outweighs a stated prior (Gupta et al., 2025) and in-context labels override a model's own priors (Min et al., 2022; Wei et al., 2023), that learning curves approach a Bayesian reference (Akata et al., 2026; Arora et al., 2025), and that text given alongside numbers shapes numerical predictions (Requeima et al., 2024; Liu et al., 2025). We add a description with a stated specificity and reliability to such a test.

3. Problem Setup

Tasks and demonstrations. A task is a weight vector $w \in \mathbb{R}^d$. A demonstration is an input $x \sim \mathcal{N}(0, I_d)$ with a label $y \sim \mathcal{N}(w^\top x, 1)$; we measure labels in units of their noise. The baseline prior over tasks is $w \sim \mathcal{N}(0, a_0 I_d)$, so a_0 is the prior signal-to-noise ratio. The numerical results of §§4–6 use $a_0 = 10$.

Descriptions. A description is a vector $m \in \mathbb{R}^d$, governed by two parameters, r and p, that we interpret below. A latent indicator $Z \sim \text{Bernoulli}(p)$, independent of m, says whether the description is *relevant* ($Z = 1$) or *irrelevant* ($Z = 0$):

$$\begin{aligned} m &\sim \mathcal{N}(0, (1-r)\, a_0 I_d) \\ w \mid m, Z = 1 &\sim \mathcal{N}(m, r a_0 I_d) \\ w \mid m, Z = 0 &\sim \mathcal{N}(0, a_0 I_d) \end{aligned}$$

In words, a relevant description says the task lies near m, within a variance ra_0 in every direction; an irrelevant one is unrelated to the task. Either way, w marginally has the baseline prior. Marginalizing over Z, the prior for w given m is the robust mixture $p\,\mathcal{N}(m, ra_0 I_d) + (1-p)\,\mathcal{N}(0, a_0 I_d)$ (Schmidli et al., 2014). The parameter $r \in (0, 1)$ is the fraction of the prior variance that a relevant description leaves, and it sets the description's **specificity**: the smaller r, the more specific the description, from one that only loosely indicates the task (r near 1) to one that all but states it (r near 0). The parameter p is the description's **reliability**. We call a description with $p = 1$ *reliable*, and write $c = 1/(ra_0)$ for the precision of a relevant description's prior, in units of demonstrations. We call a description *loose* when c is small next to d and *specific* when c exceeds d; at $d = 16$ and $a_0 = 10$, $r = 0.1$ ($c = 1$) is loose and $r = 0.001$ ($c = 100$) is specific.

Error and ESS. A predictor sees a prompt D (the description, n demonstrations, and a query x) and predicts the query's label. As Garg et al. (2022) do, we score its prediction $\hat{y}$ by its error, the squared distance from the noiseless label $w^\top x$:

$$R = \mathbb{E}\big[(\hat{y} - w^\top x)^2\big], \tag{1}$$

the expected squared error less the noise variance. Let $R^{\text{desc}}(n)$ be a predictor's error with the description and n demonstrations, and $R^{\text{demo}}(n)$ its error with n demonstrations alone. In §§4–5 the predictor is Bayes-optimal: it predicts the posterior mean of $w^\top x$, so its error is the expected posterior variance of $w^\top x$, and $R^{\text{desc}}_{p=1}(n)$ is the value of $R^{\text{desc}}(n)$ for a reliable description. The ESS counts the further demonstrations that a predictor without the description would need to match the predictor with it.

Definition 1. Given n demonstrations, the effective sample size (ESS) of a description is $\text{ESS}_n = n^* - n$, where n^* is the smallest $t \geq 0$ with $R^{\text{demo}}(t) = R^{\text{desc}}(n)$, with R^{demo} interpolated between the numbers of demonstrations at which it is known. If $R^{\text{demo}}(t) < R^{\text{desc}}(n)$ for every t, we write $\text{ESS}_n \leq -n$: the description does more harm than all n demonstrations do good.

If $R^{\text{desc}}(n)$ is below R^{demo} at every t measured, up to $t_{\max}$, we write $\text{ESS}_n > t_{\max} - n$, and if above it at the smallest, $t_{\min}$, $\text{ESS}_n \le t_{\min} - n$; §7 estimates ESS_n from a sparse grid of t. The errors average over tasks, descriptions, and demonstrations, so the ESS belongs to a kind of description, set by r and p, not to one value of m. For the Bayes-optimal predictor the ESS is never negative: conditioning on the description cannot raise the expected posterior variance, so $R^{\text{desc}}(n) \le R^{\text{demo}}(n)$, and R^{demo} decreases in n (Appendix A.1). For any other learner the ESS is read from that learner's own two error curves (§§6–7), so it measures what the description is worth to that learner, and it is negative when the description hurts.

4. Specificity: Reliable Descriptions

A reliable description narrows the prior over the task; Proposition 1 counts the demonstrations that would narrow it as much. Write $W_n = X_n^\top X_n$, with the n demonstration inputs as the rows of X_n. Without the description the posterior covariance of w is $(I/a_0 + W_n)^{-1}$, and a reliable description replaces the baseline prior's precision $1/a_0$ with its own, c. Averaged over the query, the error is the trace of the posterior covariance:

$$R^{\text{demo}}(n) = \mathbb{E}\,\text{tr}(I/a_0 + W_n)^{-1},$$
$$R^{\text{desc}}_{p=1}(n) = \mathbb{E}\,\text{tr}(cI + W_n)^{-1}.$$

Proposition 1 (ESS of a reliable description)**.** *Let* $p = 1$.

(i) (Loose descriptions, worth about $(1-r)d$*)* $\text{ESS}_0 \ge (1-r)\,d$*, and* $\text{ESS}_0 \to (1-r)\,d$ *as* $a_0 \to \infty$ *with* r *fixed.*

(ii) (Specific descriptions, worth $c+d+1$*) If* c *is a whole number,* $\text{ESS}_0 \to c+d+1$ *as* $a_0 \to \infty$ *with* c *fixed.*

Appendix A.1 proves it from two standard facts. With $n < d$ demonstrations the data say nothing about $d-n$ directions, so $R^{\text{demo}}(n) \ge (d-n)\,a_0$. With a flat baseline prior and $n \ge d+2$, $R^{\text{demo}}(n) = d/(n-d-1)$, the trace of the mean of an inverse Wishart matrix (Muirhead, 1982). A reliable description leaves $R^{\text{desc}}_{p=1}(0) = d/c = ra_0 d$, and the ESS is where these curves fall to that value.

The ESS has two parts. The first, the *dimension term*, counts dimensions: each of the first d demonstrations removes the uncertainty in one direction, so a loose description, which removes a fraction $1-r$ of the prior variance in every direction, is worth about $(1-r)d$ of them, and a specific one about d. A specific description is also worth c, the ESS that Morita et al. (2008) give a normal prior, the ratio of the observation variance to the prior variance. Their ESS matches information, so a description that raises the prior precision from $1/a_0$ to c is worth $(1-r)\,c$ at every n. Ours matches errors, and n demonstrations leave the error $d/(n-d-1)$, not the d/n of their expected information; the dimension term is that gap, $d+1$ for a specific description, and it fades as demonstrations accumulate (Corollary 4(ii)).

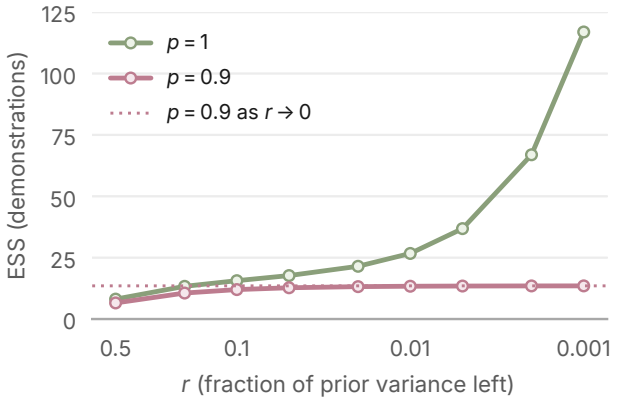


Figure 2. **Specificity raises the ESS of a reliable description without bound; one that may be irrelevant stays below a bound.** Bayes-optimal ESS of a description with no demonstrations in hand, against r, the fraction of the prior variance it leaves, for a reliable description ($p = 1$) and one that is relevant with probability 0.9. Dotted: the limit of the $p = 0.9$ ESS as $r \to 0$. $d = 16$, $a_0 = 10$; eight batches of 4,000 prompts, standard errors below 0.02 demonstrations.

At $d = 16$ and $a_0 = 10$, a description that leaves a tenth of the prior variance ($r = 0.1$, $c = 1$) is worth 15.6 demonstrations, against $(1-r)d = 14.4$, and one that leaves a thousandth ($r = 0.001$, $c = 100$) is worth 117, matching $c + d + 1$ (Figure 2). At $r = 0.1$, 100 demonstrations in hand lower the ESS from 15.6 to 1.1.

Descriptions of part of the task. A description can be specific about some aspects of a task and silent about others. Let a reliable description leave the fraction r of the prior variance in $k < d$ directions and all of it in the other $d-k$, so that its prior precision is c in those k directions and $1/a_0$ in the rest.

Proposition 2 (ESS of a partial description)**.** *Let* $p = 1$ *and* $k < d$.

(i) (Without demonstrations, worth about k*) As* $a_0 \to \infty$, $\text{ESS}_0 \to (1-r)\,k$ *with* r *fixed and* $\text{ESS}_0 \to k$ *with* c *fixed.*

(ii) (With many demonstrations, worth $(1-r)ck/d$*) As* $n \to \infty$, $\text{ESS}_n \to (1-r)\,c\,k/d$.

Without demonstrations, a partial description is worth the directions it pins down and its precision adds nothing, because the error left in the directions it is silent about dwarfs what precision removes (Appendix A.3); the $c + d + 1$ of Proposition 1(ii) needs a description of every direction. Once demonstrations have resolved the other directions, the description is worth the precision it adds, $k(1-r)c$, in units of the information d that one demonstration carries.

When that exceeds k, about when $(1-r)c > d$, a partial description gains value as demonstrations accumulate: at $d = 16$, $a_0 = 10$, and $r = 0.001$, one of half the directions is worth 8.1 demonstrations with none in hand and 36 with 100, while a complete one falls from 117 to 109 (Table 1). Demonstrations substitute for a complete description and complement a partial one.

5. Reliability: Descriptions That May Be Irrelevant

Proposition 3 gives the price of not knowing whether a description is relevant, and the error in closed form when no demonstrations are in hand. When $p < 1$ the Bayes-optimal prediction mixes two predictions, $\mu_1^\top x$, which takes the description as relevant, and $\mu_0^\top x$, which ignores it (μ_1 and μ_0 are the posterior means of w under the two readings), weighted by the posterior probability π_n that the description is relevant.

Proposition 3 (Error of a description that may be irrelevant)**.** *For every $n \geq 0$,*

$$R^{\text{desc}}(n) \;=\; p\, R^{\text{desc}}_{p=1}(n) + (1-p)\, R^{\text{demo}}(n) + \mathcal{V}_n, \quad (2)$$

where $\mathcal{V}_n = \mathbb{E}\big[\pi_n(1-\pi_n)\big((\mu_1-\mu_0)^\top x\big)^2\big] \geq 0$. With no demonstrations in hand,

$$R^{\text{desc}}(0) = \big(1 - p^2(1-r)\big)\, a_0 d. \quad (3)$$

The decomposition (2) is the law of total variance over Z (Appendix A.2). The first two terms are the error of a *cheating* predictor, one that is told whether the description is relevant: with probability p it holds a reliable description, and with probability $1-p$ it holds only the demonstrations. To that predictor, a description that may be irrelevant is worth what a reliable one is worth when it is relevant and nothing when it is not.

The last term, $\mathcal{V}_n$, is the price of not being told: the variance of the prediction between the two readings of the description. It is small when the readings agree or when the demonstrations make one reading far likelier than the other: with no demonstrations it grows as a more specific description pulls the readings apart, but with demonstrations a specific description is also easier to tell from an irrelevant one. With no demonstrations, (3) puts the two predictors side by side: of the error $a_0 d$ that the baseline prior leaves, the cheating predictor removes a fraction $p(1-r)$ and the Bayes-optimal predictor $p^2(1-r)$, so not being told costs $p(1-p)(1-r)\, a_0 d$.

Corollary 4 (ESS of a description that may be irrelevant)**.**

(i) (Without demonstrations) If $p < 1$, then $\text{ESS}_0 \leq n_p^{\max}$, where $n_p^{\max}$, which depends on d, a_0, and p but not on r, solves $R^{\text{demo}}(n_p^{\max}) = (1-p)\, R^{\text{demo}}(0)$ for the interpolated R^{demo}. Also $\text{ESS}_0 \geq p^2(1-r)\, d$, and as $a_0 \to \infty$ with r fixed, $n_p^{\max} \to p\, d$ and $\text{ESS}_0 \to p^2(1-r)\, d$.

(ii) (With many demonstrations) As $n \to \infty$, $\text{ESS}_n \to (1-r)\, c$ if $p = 1$, and if $p < 1$,

$$\limsup_{n\to\infty} \text{ESS}_n \leq p\,(1-r)\, c.$$

Without demonstrations the ESS is bounded. Refining a description from $r = 0.1$ to $r = 0.001$ raises its ESS from 15.6 to 117 if it is reliable, but only from 12.0 to 13.5 if it is relevant with probability 0.9. However specific the description, the average error cannot fall below the share contributed by irrelevant descriptions, $(1-p)R^{\text{demo}}(0)$, and Corollary 4(i) turns that floor into a bound on the ESS: $n_p^{\max} = 15.6$ demonstrations at $d = 16$, $a_0 = 10$, and $p = 0.9$, the ESS of a reliable description that leaves a fraction $1-p$ of the prior variance, which leaves the same error. The bound holds even for a predictor told whether the description is relevant, so it comes from irrelevant descriptions being worth nothing, not from uncertainty about relevance. That uncertainty lowers the ESS further, from about pd to about $p^2 d$ at a high signal-to-noise ratio: as $r \to 0$ the ESS approaches 13.5 (dotted in Figure 2), against $p^2 d = 13.0$.

With many demonstrations, the ESS is at most p times a reliable description's. Demonstrations shrink the error in the irrelevant case, and the ESS can grow. In the limit a reliable description is worth $(1-r)\, c$, the precision it adds to the baseline prior, and one that may be irrelevant at most p times as much: with 100 demonstrations in hand, the $r = 0.001$ description is worth 109 if reliable and 86.6 at $p = 0.9$, against limits of 99.9 and at most 89.9.

Full trust can be worse than no description. A *fully trusting* predictor treats every description as relevant, predicting as if $p = 1$. With no demonstrations its error is

$$R^{\text{trust}}(0) = \big(1 - (2p-1)(1-r)\big)\, a_0 d \quad (4)$$

(Appendix A.4): it removes a fraction $(2p-1)(1-r)$ of the prior error, against $p^2(1-r)$ for the Bayes-optimal predictor, and does worse than ignoring the description only when $p < 1/2$. Demonstrations do not rescue it: they show whether a specific description is relevant, but it keeps predicting near m, and as $r \to 0$ its error tends to $2(1-p)\, a_0 d$ at every n. At $d = 16$, $a_0 = 10$, and $p = 0.9$, it is worse than no description from 15 demonstrations on.

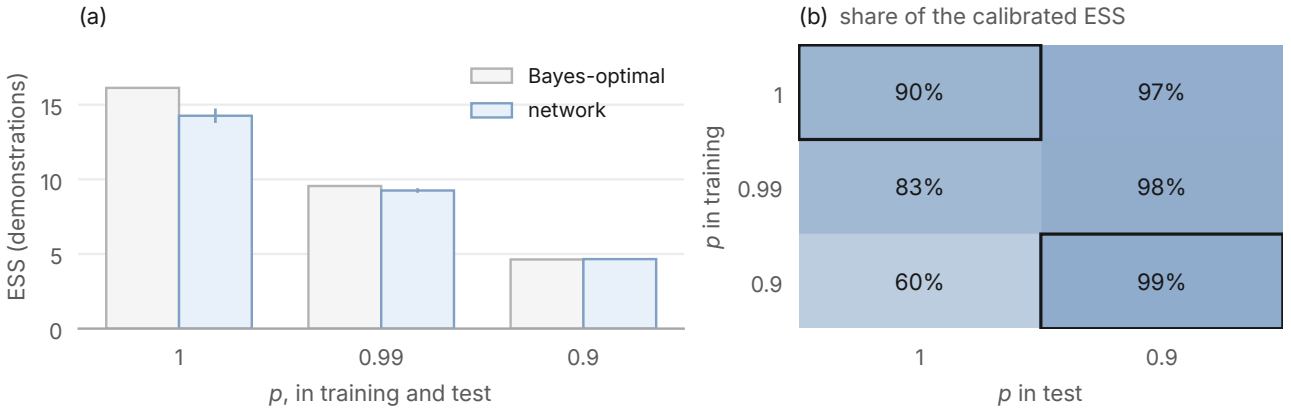


Figure 3. **Networks come close to the Bayes-optimal ESS and take their trust from training.** Specific description ($r = 0.01$), $d = 5$, $a_0 = 10$, no demonstrations; the network's ESS is its mean over three training seeds, with error bars of $\pm$ one standard deviation across them. (a) Trained and tested at the same reliability p: the Bayes-optimal ESS on the same prompts and the network's, read from the network's own no-description curve. (b) Trained at one reliability and tested at another: the network's ESS as a share of the calibrated predictor's; outlined, training matches test. Table 2 (Appendix B) gives every value, including those for the loose description ($r = 0.1$).

6. Meta-Trained Transformers

We meta-train Transformers on the setting of §3, from which they must learn how specific and how reliable descriptions are, and ask how close they come to the Bayes-optimal ESS and what sets their trust in a description.

Setup. Prompts use the prefix embedding of Huang & Ge (2025): an optional description token carrying m, up to 20 demonstration tokens, and a query token. The network has the size of Garg et al. (2022)'s (12 layers, 8 heads, width 256) and their optimizer, but is a bidirectional Transformer without positional encoding (Appendix B). To keep training short we use $d = 5$, with $a_0 = 10$ as before. The network outputs a mean and a log variance for the query's label and is trained with log loss, as in Zhu et al. (2026); the mean that minimizes this loss is the posterior mean, the Bayes-optimal prediction under squared error, and we score the network's mean by its squared error (1). The description token carries only m, and half the training prompts omit it, so each network must learn r and p from its training distribution. We train a network for each specificity ($r = 0.1$, loose; $r = 0.01$, specific) and reliability ($p \in \{1, 0.99, 0.9\}$), with three training seeds for the specific description and two for the loose one.

Evaluation. We test each network on 20,000 fresh prompts at every n from 0 to 20, with and without a description, at its training reliability and at $p = 1$ and 0.9 (later seeds also at 0.99; Table 2). Its ESS follows Definition 1 with its own error curves (without a description, its error is close to the Bayes-optimal predictor's; Appendix B). We compare the network's ESS with the *trained* Bayes-optimal predictor, which uses the reliability the network was trained on, and the *calibrated* one, which uses that of the test prompts. A network's *implied trust* is the q whose Bayes-optimal prediction with no demonstrations, $q\, m^\top x$, is closest to the network's in squared error, the regression coefficient of its prediction on $m^\top x$ (Table 2).

Networks come close to the Bayes-optimal ESS. Trained and tested at the same reliability (Figure 3a), for the specific description the network's ESS is 14.3 ± 0.5 (mean $\pm$ standard deviation over three seeds, on the same prompts) against a Bayes-optimal 16.1 on them at $p = 1$, 9.3 ± 0.2 against 9.6 at $p = 0.99$, and 4.66 ± 0.01 against 4.63 at $p = 0.9$ (on 200,000 prompts, 15.9, near the $c + d + 1 = 16$ of Proposition 1(ii)); for the loose one, every setting is within 0.1 of the Bayes-optimal ESS (Table 2).

Training sets a network's trust. With no demonstrations the prompt says nothing about relevance (Appendix A.2, Step 1), so any learner must take its trust from training; what the networks show is how much (Figure 3b). Networks trained at $p = 0.9$ keep hedging, at an implied trust of 0.87 to 0.89 across three seeds, even when every test description is relevant, and reach an ESS of 9.5 ± 0.4 where the calibrated predictor reaches 15.9 and the trained one 10.8. Networks trained on specific descriptions at $p = 1$ or 0.99 treat every description as relevant when no demonstrations are shown (implied trust 0.95 to 1.02). Demonstrations separate them: when one test description in ten is irrelevant, the description lowers the $p = 0.99$ networks' error at every n, as it does the Bayes-optimal predictor's, while the $p = 1$ networks trust fully. As §5 predicts, this costs little with no demonstrations (4.58 ± 0.03 against 4.70), but in every seed, from 6 demonstrations on, their error is higher with the description than without, as the fully trusting Bayes-optimal predictor's is.

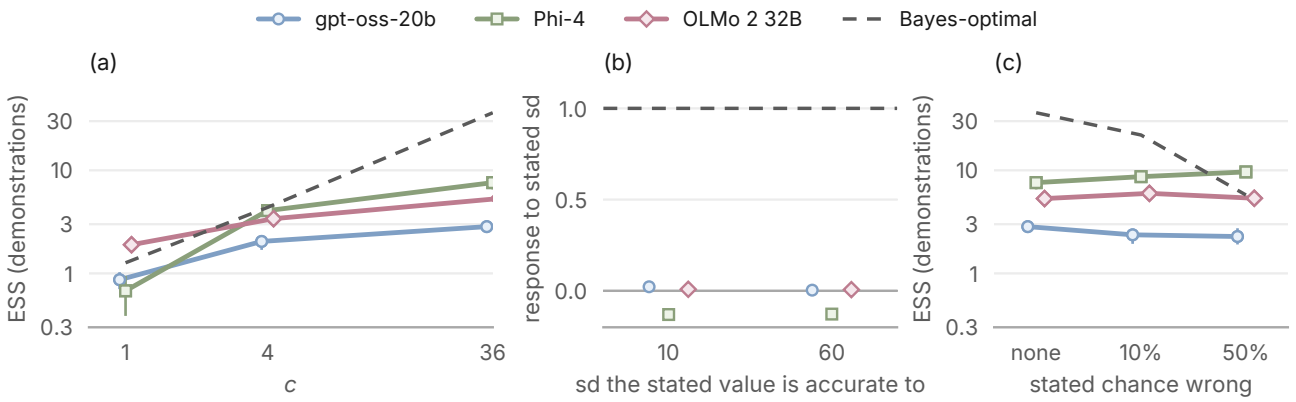


Figure 4. **Language models barely adjust to a description's stated specificity or reliability.** gpt-oss-20b, Phi-4, and OLMo 2 32B, read from their first-token distribution and answering without reasoning (the other three are in Table 3, Appendix C), against the Bayes-optimal predictor (dashed), for a relevant description; $\sigma = 60$, 2,000 tasks; error bars, 95% bootstrap intervals over tasks; in (a) and (c) the ESS is the median over $n = 1$, 2, and 4 demonstrations shown. (a) ESS of the descriptions with Bayes-optimal ESS $c = 1$, 4, and 36. (b) The specificity control (Table 7): the change in prediction when only the stated standard deviation goes from 60 to 10, as a share of the Bayes-optimal change over $n = 1$, 2, and 4, for a stated value accurate to standard deviation 10 or 60. (c) ESS of the $c = 36$ description with no stated chance of being wrong, or a stated 10% or 50% chance; the description is in fact relevant, and the Bayes-optimal predictor takes the stated chance at face value.

7. Pretrained Language Models

The networks learn a description's specificity and reliability from training; a pretrained language model can learn them only from the prompt. We therefore state both in the prompt, together with the noise level, and compare the description's ESS for the model with its ESS for a predictor that is Bayes-optimal given what the prompt states. The design follows Gupta et al. (2025) and Akata et al. (2026): a stated prior is a description whose Bayes-optimal ESS is known exactly, so the model is measured against a known optimum.

Setting. We use the simplest case of §3: one weight ($d = 1$) and a fixed input $x = 1$. A task is a hidden mean $w \sim \mathcal{N}(500, 100^2)$, and a demonstration is a number $y \sim \mathcal{N}(w, \sigma^2)$, rounded to an integer. A description states a value m and a standard deviation τ, with $w \mid m \sim \mathcal{N}(m, \tau^2)$: the description of §3 with $r = \tau^2/100^2$. We set $\sigma = 60$ and $\tau = 60$, 30, or 10, and show $n = 0$, 1, 2, 4, …, 64 demonstrations; $\sigma = 30$ with $\tau = 30$, 15, or 5 serves as a check. An *irrelevant* description states the value that a relevant description of an independently drawn task would state. A *stated reliability* adds a sentence giving a 10% or 50% chance that the description is wrong ($p = 0.9$ or 0.5, in words) and keeps the stated value and the demonstrations. A *specificity control* states the value drawn for $\tau = 10$, or for $\tau = 60$, with each of $\tau = 60$, 30, and 10, on the same tasks and demonstrations, so that only the stated specificity changes. The description is one sentence of the prompt, such as "Its mean was drawn from a normal distribution with mean 510 and standard deviation 30" (Appendix C).

Bayes-optimal predictor. The Bayes-optimal predictor here knows only what the prompt says: the noise σ and the description, with the mean otherwise uniform over the three-digit numbers (Appendix C gives the ESS under the prior the tasks are drawn from). Given a relevant description and n demonstrations whose mean is $\bar{y}$, it predicts $(c\,m + n\,\bar{y})/(c + n)$ with $c = \sigma^2/\tau^2$, a weight of $c/(c + n)$ on the stated value. Its squared error is $\sigma^2/(c + n)$ with the description and σ^2/n without, so the description's Bayes-optimal ESS is c at every n, up to the truncation of that uniform prior. It takes an irrelevant description as relevant unless told otherwise, so a model can do better than it. Our values of τ give $c = 1$, 4, and 36 at both noise levels, and we name the descriptions by c.

Models. We use six open chat models. gpt-oss-20b (OpenAI, 2025), Phi-4 (14B; Abdin et al., 2024), and OLMo 2 32B Instruct (Team OLMo et al., 2025) encode every three-digit integer as one token, so the reply's first-token distribution, restricted to the 900 number tokens and renormalized, is the model's distribution over the next number, and its prediction is that distribution's mean. Qwen3.5 9B (Qwen Team, 2026), Gemma 4 12B, and Gemma 4 31B (Gemma Team, 2026) split numbers into digits, so their prediction is the number they write under greedy decoding; on gpt-oss-20b the two readouts give weights within 0.21 of each other (Appendix C). Three others begin fewer than three quarters of their replies with a number and are not reported. The prompt is the user turn of the chat template (Appendix C).

Measurement. A model's error is the root-mean-square distance from the hidden mean over 2,000 tasks. We read

Definition 1, with squared error, off the model's own no-description curve at $n = 1, 2, 4, \ldots, 64$, interpolated in $\log n$, at its first crossing, and summarize it by its median over $n = 1, 2$, and 4 with 95% bootstrap intervals over tasks (every n and other readings of the curve, which move no model's median by more than 1.4, are in Appendix C). Read this way, the Bayes-optimal ESS is 1.3, 4.4, and 36. Where a model's own no-description curve rises, we also read the *reference ESS* off the Bayes-optimal one. A model's *weight* on the stated value is the coefficient on m in a regression of its prediction on m, the demonstrations' mean, and a constant; it is $c/(c+n)$ for the Bayes-optimal predictor.

A specific description is worth a small fraction of its Bayes-optimal ESS. Across the six models, the $c = 36$ description is worth a median of 1.4 to 7.6 demonstrations against a Bayes-optimal 36; the $c = 1$ and $c = 4$ descriptions are worth 0.0 to 1.9 and 0.2 to 4.1, against 1.3 and 4.4 (Figure 4a; Table 3). The ESS rises with c mainly because a more specific description states a value closer to the truth: in the specificity control, with the value accurate to $\tau = 10$, gpt-oss-20b's ESS is 2.7, 2.6, and 2.8 whether the prompt states $\tau = 60$, 30, or 10, against a Bayes-optimal 2.5, 13, and 36 (Table 7).

The weight follows the stated specificity weakly or not at all. The specificity control measures this directly. Changing only the stated standard deviation from 60 to 10 moves the Bayes-optimal weight by 0.47 at one demonstration; across $n = 1$, 2, and 4, gpt-oss-20b and OLMo 2 32B move by at most 0.02 of the Bayes-optimal change, Phi-4 by -0.13, away from a value stated as more precise, and the three models read only from their written answer by 0.12 to 0.16 (Qwen3.5 9B) up to 0.28 to 0.38 (Gemma 4 31B, mostly at one demonstration; Figure 4b, Table 7). In the main design the first three models' weight spreads across c by at most 0.16, against 0.34 to 0.71 (Table 13). So the models under-use a specific description and over-use a loose one: at 64 demonstrations gpt-oss-20b still puts weight 0.23 on the $c = 1$ value, against the Bayes-optimal 0.02, and that description raises its error (ESS -57).

Models discount a contradicted description. An irrelevant $c = 36$ description has an ESS of -0.4 or less, but at $n = 4$ the models' errors (35 to 64) stay far below the 124 of the Bayes-optimal predictor, which takes it as relevant.

A stated chance of being wrong does not move the ESS in proportion. Stating a 10% or 50% chance that the description is wrong leaves the ESS of gpt-oss-20b and OLMo 2 32B about where it was and raises Phi-4's (from 7.6 to 8.7 and 9.7; Figure 4c); Qwen3.5 9B and Gemma 4 31B lower it about equally for 10% and 50%, and Gemma 4 12B's falls for 10% but rises for 50% (Table 3). The Bayes-optimal ESS falls to 22 and 5.8 (actual unreliability: Table 5).

With reasoning, the weight follows the stated specificity. We reran these conditions on 200 tasks with two models that reason first: gpt-oss-20b at low effort with 4,096 new tokens, so that 99.9% of replies finish, and Claude Haiku 5.5 through the Claude Code command line (Appendix C). Both now move with the stated standard deviation as the Bayes-optimal predictor does, by 1.0 and 0.92 of its change in the control (Table 7). Haiku 5.5 also comes close to the Bayes-optimal ESS, 1.1, 3.3, and 28 against 1.0, 3.5, and 35 (Table 8). It hedges when told of a 50% chance of being wrong (weight 0.55 against 0.58) but not of a 10% one, and gpt-oss-20b barely hedges at all. GPT-5.6-Luna, reasoning in the Codex agent, matches the Bayes-optimal weight to within 0.03 and its $c = 36$ ESS (32 against 33; Appendix C). gpt-oss-20b gets the weight right, but about 2% of its answers are more than 40 from the Bayes-optimal prediction, from slips such as swapping the two variances, so its $c = 36$ description has a reference ESS of 24 $[12, 35]$; read off its own error curve, which rises after 16 demonstrations, its ESS exceeds the grid and says nothing. A fitted weight hides such errors; the reference ESS does not.

8. Conclusion

For the Bayes-optimal predictor in in-context linear regression, a reliable description's ESS grows with the task dimension and with its specificity, a description of only part of the task is worth about the number of directions it covers until demonstrations resolve the rest, and one that may be irrelevant has a bounded ESS without demonstrations and can hurt a learner that trusts it fully once they are in hand. Meta-trained networks come close to the Bayes-optimal ESS and take their trust from training. Pretrained language models answering without reasoning respond to a description's stated specificity far less than the Bayes-optimal predictor, even when nothing else in the prompt changes; models that reason follow it, and two come close to the Bayes-optimal ESS, though a few large errors can cost a reasoning model much of its ESS while leaving its fitted weight intact.

Limitations. The analysis assumes a linear-Gaussian task and the robust mixture as the process generating descriptions, so a relevant description errs only within its stated spread and an irrelevant one carries no information; a real description can be partly right or more confident than it is accurate. The networks are trained at $d = 5$, and partial descriptions are not tested. The language-model test covers the simplest case ($d = 1$, with no dimension term), with a stated numeric prior rather than an instruction; three open models are read from a greedy written answer, and the reasoning runs use 200 tasks.

Impact Statement

This paper gives task descriptions a value in demonstrations and tests whether learners rely on them accordingly, which bears on how descriptions and demonstrations are combined in prompts. We see no societal consequences to discuss.

References

Abdin, M., Aneja, J., Behl, H., et al. Phi-4 technical report. Technical report, Microsoft, 2024. arXiv:2412.08905.

Akata, E., Voudouris, K., Fortuin, V., and Schulz, E. In-context function learning in large language models. In *AISTATS*, 2026.

Akyürek, E., Schuurmans, D., Andreas, J., Ma, T., and Zhou, D. What learning algorithm is in-context learning? Investigations with linear models. In *ICLR*, 2023.

Arora, A., Jurafsky, D., Potts, C., and Goodman, N. D. Bayesian scaling laws for in-context learning. In *COLM*, 2025.

Brown, T. B. et al. Language models are few-shot learners. In *NeurIPS*, 2020.

Chatlatanagulchai, W., Thonglek, K., Reid, B., Kashiwa, Y., Leelaprute, P., Rungsawang, A., Manaskasemsak, B., and Iida, H. On the use of agentic coding manifests: an empirical study of Claude Code. In *International Conference on Product-Focused Software Process Improvement (PROFES)*, 2025.

Cook, J., Sapora, S., Ahmadian, A., Khan, A., Rocktäschel, T., Foerster, J., and Ruis, L. Programming by backprop: an instruction is worth 100 examples when finetuning LLMs. In *ICLR*, 2026.

Evans, M. and Moshonov, H. Checking for prior-data conflict. *Bayesian Analysis*, 1(4):893–914, 2006.

Garg, S., Tsipras, D., Liang, P., and Valiant, G. What can transformers learn in-context? A case study of simple function classes. In *NeurIPS*, 2022.

Gemma Team. Gemma 4. https://huggingface.co/google/gemma-4-31B-it, 2026. Model card.

Gloaguen, T., Mündler-Sasahara, N., Müller, M. N., Raychev, V., and Vechev, M. Evaluating AGENTS.md: are repository-level context files helpful for coding agents? *arXiv:2602.11988*, 2026.

Gupta, R., Corona, R., Ge, J., Wang, E., Klein, D., Darrell, T., and Chan, D. M. Enough coin flips can make LLMs act Bayesian. In *ACL*, 2025.

Honda, U., Murakami, S., and Zhang, P. Distilling many-shot in-context learning into a cheat sheet. In *Findings of EMNLP*, 2025.

Huang, R. and Ge, R. Task descriptors help transformers learn linear models in-context. In *ICLR*, 2025.

Le Scao, T. and Rush, A. M. How many data points is a prompt worth? In *NAACL*, 2021.

Lin, Z., Bharti, S. K., and Lee, K. In-context learning with hypothesis-class guidance. *arXiv:2502.19787*, 2025.

Liu, T. J. B., Boullé, N., Sarfati, R., and Earls, C. J. Density estimation with LLMs: a geometric investigation of in-context learning trajectories. In *ICLR*, 2025.

Min, S., Lyu, X., Holtzman, A., Artetxe, M., Lewis, M., Hajishirzi, H., and Zettlemoyer, L. Rethinking the role of demonstrations: What makes in-context learning work? In *EMNLP*, 2022.

MiniCPM Team. MiniCPM5-2B. https://huggingface.co/openbmb/MiniCPM5-2B, 2026. Model card; the series is described in *MiniCPM4: Ultra-Efficient LLMs on End Devices*, arXiv:2506.07900.

Morita, S., Thall, P. F., and Müller, P. Determining the effective sample size of a parametric prior. *Biometrics*, 64(2):595–602, 2008.

Muirhead, R. J. *Aspects of Multivariate Statistical Theory*. Wiley, 1982.

Müller, S., Hollmann, N., Pineda Arango, S., Grabocka, J., and Hutter, F. Transformers can do Bayesian inference. In *ICLR*, 2022.

Neuenschwander, B., Weber, S., Schmidli, H., and O'Hagan, A. Predictively consistent prior effective sample sizes. *Biometrics*, 76(2):578–587, 2020.

OpenAI. gpt-oss-120b & gpt-oss-20b model card. Technical report, OpenAI, 2025. arXiv:2508.10925.

Panwar, M., Ahuja, K., and Goyal, N. In-context learning through the Bayesian prism. In *ICLR*, 2024.

Qwen Team. Qwen3.5-9B. https://huggingface.co/Qwen/Qwen3.5-9B, 2026. Model card.

Raventós, A., Paul, M., Chen, F., and Ganguli, S. Pretraining task diversity and the emergence of non-Bayesian in-context learning for regression. In *NeurIPS*, 2023.

Reimherr, M., Meng, X.-L., and Nicolae, D. L. Prior sample size extensions for assessing prior impact and prior-likelihood discordance. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 83(3):413–437, 2021.

Requeima, J., Bronskill, J., Choi, D., Turner, R. E., and Duvenaud, D. LLM processes: numerical predictive distributions conditioned on natural language. In *NeurIPS*, 2024.

Reznik, Y. A. The optimal discounting parameter of the power prior under predictive log-loss. *arXiv:2608.12159*, 2026.

Schmidli, H., Gsteiger, S., Roychoudhury, S., O'Hagan, A., Spiegelhalter, D., and Neuenschwander, B. Robust meta-analytic-predictive priors in clinical trials with historical control information. *Biometrics*, 70(4):1023–1032, 2014.

Team OLMo, Walsh, P., Soldaini, L., Groeneveld, D., Lo, K., et al. 2 OLMo 2 Furious. In *COLM*, 2025.

Tong, X., Zeng, Y., and Zhang, J. Demonstrations, CoT, and prompting: a theoretical analysis of ICL. *arXiv:2603.19611*, 2026.

von Rosen, D. Moments for the inverted Wishart distribution. *Scandinavian Journal of Statistics*, 15(2):97–109, 1988.

Webson, A. and Pavlick, E. Do prompt-based models really understand the meaning of their prompts? In *NAACL*, 2022.

Wei, J., Wei, J., Tay, Y., Tran, D., Webson, A., Lu, Y., Chen, X., Liu, H., Huang, D., Zhou, D., and Ma, T. Larger language models do in-context learning differently. *arXiv:2303.03846*, 2023.

Wenliang, L. K., Ruoss, A., Grau-Moya, J., Hutter, M., and Genewein, T. Why is prompting hard? Understanding prompts on binary sequence predictors. In *AISTATS*, 2026. PMLR 300:2134–2142.

Wiesenfarth, M. and Calderazzo, S. Quantification of prior impact in terms of effective current sample size. *Biometrics*, 76(1):326–336, 2020.

Xie, S. M., Raghunathan, A., Liang, P., and Ma, T. An explanation of in-context learning as implicit Bayesian inference. In *ICLR*, 2022.

Zhu, Q., Oermann, E. K., and Cho, K. Multi-task Bayesian in-context learning. In *ICML*, 2026. arXiv:2606.20538.

A. Proofs

A.1. Proof of Proposition 1 (Reliable Descriptions)

Notation and standard facts. Throughout, $\varepsilon = 1/a_0$, $X_n \in \mathbb{R}^{n\times d}$ holds the demonstration inputs as rows, $W_n = X_n^\top X_n$, and for $\kappa > 0$

$$R^\kappa(n) = \mathbb{E}\,\mathrm{tr}(\kappa I + W_n)^{-1},$$

the error with n demonstrations under a Gaussian prior of precision κ in every direction, so that $R^{\mathrm{demo}} = R^\varepsilon$ and $R^{\mathrm{desc}}_{p=1} = R^c$. These are errors because the Bayes-optimal prediction is the posterior mean of $w^\top x$ and the query is independent of the rest of the prompt with $\mathbb{E}[xx^\top] = I$, so the error $\mathbb{E}[x^\top \Sigma x]$ is $\mathbb{E}\,\mathrm{tr}\,\Sigma$ for the posterior covariance Σ. We use three facts. (W1) For $n < d$, W_n has rank n almost surely, so $(\kappa I + W_n)^{-1}$ has the eigenvalue $1/\kappa$ with multiplicity $d - n$. (W2) For $n \geq d$ and a fixed unit vector e, $1/(e^\top W_n^{-1} e) \sim \chi^2_{n-d+1}$ (Muirhead, 1982, Theorem 3.2.12), so $\mathbb{E}\,\mathrm{tr}\,W_n^{-1} = d/(n-d-1)$ for $n \geq d+2$ and $\mathbb{E}\,\mathrm{tr}\,W_n^{-1} = \infty$ for $n \in \{d, d+1\}$. (W3) As $n \to \infty$ with d fixed, $\mathbb{E}\,\mathrm{tr}\,W_n^{-2} = d/n^2 + O(n^{-3})$ and $\mathbb{E}\,\mathrm{tr}\,W_n^{-3} = O(n^{-3})$ (von Rosen, 1988). By the Sherman–Morrison formula each new demonstration lowers $(\kappa I + W_n)^{-1}$ by a nonzero positive semidefinite matrix, so R^κ is strictly decreasing in n, and by (W2) it tends to 0.

Proof of Proposition 1. With the description and no demonstrations, $R^{\mathrm{desc}}_{p=1}(0) = d/c = ra_0 d$.

(i). By (W1), $R^\varepsilon(n) \geq (d-n)\,a_0$ for $n < d$, and trivially for $n \geq d$, so the interpolated R^{demo} satisfies $R^{\mathrm{demo}}(t) \geq (d-t)\,a_0$ at every real $t \geq 0$. For $t < (1-r)d$ this exceeds $ra_0 d$, so $\mathrm{ESS}_0 \geq (1-r)d$. As $a_0 \to \infty$, $R^\varepsilon(n)/a_0 = \mathbb{E}\,\mathrm{tr}(I + a_0 W_n)^{-1} \to (d-n)^+$ by dominated convergence, because each eigenvalue $1/(1+a_0\lambda)$ lies in $(0,1]$ and tends to 1 if $\lambda = 0$ and to 0 otherwise. So the interpolated $R^{\mathrm{demo}}(t)/a_0$ tends to $(d-t)^+$ at every t, a curve that falls through rd only at $t = (1-r)d$; as R^{demo} is decreasing, $\mathrm{ESS}_0 \to (1-r)d$.

(ii). With c fixed, $R^{\mathrm{desc}}_{p=1}(0) = d/c$ does not depend on a_0, while $R^\varepsilon(n)$ increases as a_0 grows and, by monotone convergence, (W1), and (W2), tends to $+\infty$ for $n \leq d+1$ (by (W1) for $n < d$ and (W2) for $n \in \{d, d+1\}$) and to $\mathbb{E}\,\mathrm{tr}\,W_n^{-1} = d/(n-d-1)$ for $n \geq d+2$. So the interpolated curve tends to $+\infty$ below $d+2$ and to the linear interpolation of $d/(n-d-1)$ above it, a strictly decreasing limit that, for whole c, equals d/c at $n = c + d + 1$; the root that defines ESS_0 therefore increases to $c + d + 1$. □

Extra precision with many demonstrations. Corollary 4(ii) needs the error to second order in $1/n$. Since $\kappa I + W_n$ and W_n commute,

$$(\kappa I + W_n)^{-1} = W_n^{-1} - \kappa\, W_n^{-2} + \kappa^2\, W_n^{-2}(\kappa I + W_n)^{-1},$$

and the last term lies between 0 and $\kappa^2 W_n^{-3}$. With (W2) and (W3), for fixed d and κ,

$$R^\kappa(n) = \frac{d}{n-d-1} - \frac{\kappa\, d}{n^2} + O(n^{-3}). \tag{5}$$

Each further demonstration lowers the first term by about d/n^2, and a prior of precision κ lowers the error by κ times as much, so once n is large it is worth κ demonstrations.

A.2. Proofs of Proposition 3 and Corollary 4 (Descriptions That May Be Irrelevant)

Let $D_n = (x_{1:n}, y_{1:n}, m, x)$ be the prompt with the query input, and let $\pi_n = P(Z = 1 \mid D_n)$ be the posterior probability that the description is relevant.

Step 1: the two readings. Given D_n and $Z = 1$, w is Gaussian with the posterior mean μ_1 and covariance Σ_1 of a reliable description. Given D_n and $Z = 0$, m is independent of w, and w is Gaussian with the posterior mean μ_0 and covariance Σ_0 from the demonstrations alone. At $n = 0$, $\pi_0 = p$: seeing m says nothing about Z, because m has the same law $\mathcal{N}(0, (1-r)a_0 I)$ in both cases, and x is independent of Z.

Step 2: the decomposition. The Bayes-optimal prediction is $\mathbb{E}[w^\top x \mid D_n] = (\pi_n \mu_1 + (1-\pi_n)\mu_0)^\top x$, and its error is $\mathbb{E}\,\mathrm{Var}(w^\top x \mid D_n)$. By the law of total variance over Z,

$$\mathrm{Var}(w^\top x \mid D_n) = \pi_n\, x^\top \Sigma_1 x + (1-\pi_n)\, x^\top \Sigma_0 x + \pi_n(1-\pi_n)\big((\mu_1 - \mu_0)^\top x\big)^2.$$

For any function g of the prompt, $\mathbb{E}[\pi_n\, g(D_n)] = \mathbb{E}[Z\, g(D_n)] = p\,\mathbb{E}[g(D_n) \mid Z = 1]$, and given $Z = 1$ the prompt has the law of a prompt with a reliable description, so $\mathbb{E}[\pi_n\, x^\top \Sigma_1 x] = p\, R^{\mathrm{desc}}_{p=1}(n)$. In the same way $\mathbb{E}[(1-\pi_n)\, x^\top \Sigma_0 x] = (1-p)\, R^{\mathrm{demo}}(n)$. This is (2).

Step 3: no demonstrations. At $n = 0$, $\pi_0 = p$, $\mu_1 = m$, and $\mu_0 = 0$, so the Bayes-optimal prediction is $p\, m^\top x$. In each coordinate, $\mathrm{Var}(w_i) = a_0$, $\mathrm{Var}(m_i) = (1-r)\, a_0$, and $\mathrm{Cov}(w_i, m_i) = p\,(1-r)\, a_0$, because w_i is m_i plus independent noise when the description is relevant and is independent of m_i when it is not. Hence

$$\mathbb{E}(w_i - p\, m_i)^2 = a_0 - 2p^2(1-r)\, a_0 + p^2(1-r)\, a_0 = \big(1 - p^2(1-r)\big)\, a_0.$$

The query is independent of (w, m) with $\mathbb{E}[xx^\top] = I$, so the error $\mathbb{E}\|w - p\, m\|^2$ is d times this, which is (3).

Proof of Corollary 4. (i). At $n = 0$, (2) with $\mathcal{V}_0 \geq 0$ gives $R^{\mathrm{desc}}(0) \geq (1-p)\, R^{\mathrm{demo}}(0)$ for every r, because $R^{\mathrm{desc}}_{p=1}(0) \geq 0$. Since R^{demo} is strictly decreasing and tends to 0 (Appendix A.1), there is a finite $n_p^{\max}$ with $R^{\mathrm{demo}}(n_p^{\max}) = (1-p)\, R^{\mathrm{demo}}(0)$, and the root that defines ESS_0 cannot exceed it. By (3), $R^{\mathrm{desc}}(0) = \big(d - p^2(1-r)\, d\big)\, a_0$, and $(1-p)\, R^{\mathrm{demo}}(0) = (d - pd)\, a_0$. The bound $R^{\mathrm{demo}}(t) \geq (d-t)\, a_0$ and the limit $R^{\mathrm{demo}}(t)/a_0 \to (d-t)^+$ from the proof of Proposition 1(i) then give $\mathrm{ESS}_0 \geq p^2(1-r)\, d$, $\mathrm{ESS}_0 \to p^2(1-r)\, d$, and $n_p^{\max} \to pd$.

(ii). Fix d, a_0, and r; every O below is as $n \to \infty$. Let $L(n) = p\, R^c(n) + (1-p)\, R^\varepsilon(n)$ be the first two terms of (2). By (5), uniformly over integers $0 \leq t \leq T = \lceil c \rceil + 1$,

$$R^{\mathrm{demo}}(n+t) - L(n) = \frac{d}{n^2}\Big[p\,(c-\varepsilon) - t + O(1/n)\Big].$$

The bracket is linear in t, so the same holds for the interpolated R^{demo} at real $t \in [0, T]$, and its root $n + \eta_n$, at which it equals $L(n)$, has $\eta_n = p\,(c-\varepsilon) + O(1/n)$, where $c - \varepsilon = (1-r)\, c$. If $p = 1$, then $R^{\mathrm{desc}} = L$ and $\mathrm{ESS}_n = \eta_n \to (1-r)\, c$. If $p < 1$, then $R^{\mathrm{desc}}(n) = L(n) + \mathcal{V}_n \geq L(n)$ and R^{demo} is decreasing, so $\mathrm{ESS}_n \leq \eta_n$ and $\limsup_n \mathrm{ESS}_n \leq p\,(1-r)\, c$. □

A.3. Proof of Proposition 2 (Partial Descriptions)

Let $\Lambda = \mathrm{diag}(cI_k, \varepsilon I_{d-k})$, so that the error with the description is $\mathbb{E}\,\mathrm{tr}(\Lambda + W_n)^{-1}$. The law of W_n is invariant under rotations, so it does not matter which k directions the description informs.

Proof of Proposition 2. (i). Without demonstrations the error with the description is $\mathrm{tr}\,\Lambda^{-1} = (kr + d - k)\, a_0$. As in the proof of Proposition 1(i), $R^{\mathrm{demo}}(t) \geq (d-t)\, a_0$ at every $t \geq 0$, so $\mathrm{ESS}_0 \geq (1-r)k$, and $R^{\mathrm{demo}}(t)/a_0 \to (d-t)^+$, uniformly in t because the curves are monotone and the limit is continuous. With r fixed, the error with the description over a_0 is $d - (1-r)k$; with c fixed, it is $d - k + k/(c\, a_0) \to d - k$. As $k < d$, the limit curve passes through each value at a single point, $(1-r)k$ or k, with slope -1, so the root that defines ESS_0 converges to that point.

(ii). Expanding twice,

$$(\Lambda + W_n)^{-1} = W_n^{-1} - W_n^{-1}\Lambda W_n^{-1} + W_n^{-1}\Lambda W_n^{-1}\Lambda(\Lambda + W_n)^{-1}.$$

By rotation invariance, $\mathbb{E}W_n^{-2} = (\mathbb{E}\,\mathrm{tr}\, W_n^{-2}/d)\, I$, so by (W3) the middle term has expected trace $\mathrm{tr}(\Lambda)/n^2 + O(n^{-3})$. For the last term, $|\,\mathrm{tr}(MB)| \leq \|M\|\,\mathrm{tr}\, B$ for $B \succeq 0$; with $B = (\Lambda + W_n)^{-1} \preceq W_n^{-1}$ its trace is at most $\|\Lambda\|^2 \|W_n^{-1}\|^2\,\mathrm{tr}\, W_n^{-1} \leq d\,\|\Lambda\|^2\,\mathrm{tr}\, W_n^{-3}$, whose expectation is $O(n^{-3})$ by (W3). Hence

$$\mathbb{E}\,\mathrm{tr}(\Lambda + W_n)^{-1} = \frac{d}{n - d - 1} - \frac{\mathrm{tr}\,\Lambda}{n^2} + O(n^{-3}),$$

which is (5) with κd replaced by $\mathrm{tr}\,\Lambda$. The proof of Corollary 4(ii) with $p = 1$ then gives $\mathrm{ESS}_n \to (\mathrm{tr}\,\Lambda - d\varepsilon)/d = k(c-\varepsilon)/d = (1-r)\, c\, k/d$. □

The same steps give the limit $(\mathrm{tr}\,\Lambda - d\varepsilon)/d$ for a reliable description with any Gaussian prior precision Λ: with many demonstrations in hand, a description is worth the precision it adds averaged over directions.

Table 1. **A specific partial description gains value as demonstrations accumulate.** Bayes-optimal ESS of a reliable description of k of the $d = 16$ directions ($k = 16$: a complete description), $a_0 = 10$, with n demonstrations in hand; the last column of each block is the limit of Proposition 2(ii), $(1 - r)ck/d$. Eight batches of 2,000 prompts; standard errors at most 0.23.

	loose, $r = 0.1$ ($c = 1$)					specific, $r = 0.001$ ($c = 100$)				
k	$n = 0$	$n = 10$	$n = 100$	$n = 400$	$n \to \infty$	$n = 0$	$n = 10$	$n = 100$	$n = 400$	$n \to \infty$
1	0.9	0.8	0.1	0.1	0.1	1.0	1.0	3.5	5.2	6.2
4	3.6	2.9	0.3	0.2	0.2	4.0	4.2	15.5	21.7	25.0
8	7.3	5.1	0.5	0.5	0.5	8.1	9.8	36.2	46.0	50.0
12	11.1	6.7	0.8	0.7	0.7	12.4	23.4	65.1	73.0	74.9
16	15.6	8.0	1.1	0.9	0.9	117	116	109	103	99.9

A.4. Derivation of Equation (4) (The Fully Trusting Predictor)

With no demonstrations the fully trusting predictor predicts $m^\top x$. With the moments of Step 3, $\mathbb{E}(w_i - m_i)^2 = a_0 - 2p(1 - r)\, a_0 + (1 - r)\, a_0 = \big(1 - (2p - 1)(1 - r)\big)\, a_0$, and summing over the d coordinates gives (4); it exceeds $R^{\mathrm{demo}}(0) = a_0 d$ exactly when $p < 1/2$. With n demonstrations it predicts $\mu_1^\top x$ whatever Z, where $\mu_1 = m + (cI + W_n)^{-1} X_n^\top (y_{1:n} - X_n m)$. As $r \to 0$ with a_0 and n fixed, $c \to \infty$ and $\mu_1 \to m$ in mean square, so its error tends to $\mathbb{E}\|w - m\|^2$: $r a_0 d \to 0$ for a relevant description and $(2 - r)\, a_0 d \to 2 a_0 d$ for an irrelevant one, $2(1 - p)\, a_0 d$ in all. Since $R^{\mathrm{demo}}(n) \to 0$, for every $p < 1$ there is an n from which full trust does worse than no description once r is small enough.

B. The Meta-Trained Networks in Full

Table 2. Networks against the reference predictors ($d = 5$, $a_0 = 10$; 20,000 evaluation prompts per network), grouped by training reliability. "Network" is the mean ESS over the training seeds, ± the standard deviation across them (for one seed, the standard error over prompts); "trust" is also averaged over seeds. "Trained" and "calibrated" are the Bayes-optimal predictors with the training and the test reliability, shown once where they coincide. Calibrated values, and trained values where the two reliabilities coincide, use 200,000 prompts (standard errors at most 0.1); other trained values use the network's evaluation prompts, on which prompt noise alone can move them by up to 0.6, so a trained value can exceed the calibrated one. "Trust" is the network's implied trust with no demonstrations.

				ESS			
r	p_{train}	p_{test}	Seeds	Network	Trained	Calibrated	Trust
0.1	1	1	2	5.85 ± 0.03	5.89	—	0.964
0.1	1	0.99	1	5.65 ± 0.06	5.66	5.61	0.991
0.1	1	0.9	2	4.02 ± 0.03	3.98	4.10	0.964
0.01	1	1	3	14.26 ± 0.48	15.91	—	1.008
0.01	1	0.99	2	9.38 ± 0.16	9.54	8.96	1.015
0.01	1	0.9	3	4.58 ± 0.03	4.54	4.70	1.008
0.1	0.99	1	2	5.87 ± 0.01	5.88	5.89	0.963
0.1	0.99	0.99	2	5.65 ± 0.01	5.61	—	0.963
0.1	0.99	0.9	2	4.02 ± 0.03	3.99	4.10	0.963
0.01	0.99	1	3	13.28 ± 0.57	16.01	15.91	0.982
0.01	0.99	0.99	3	9.25 ± 0.15	8.96	—	0.982
0.01	0.99	0.9	3	4.60 ± 0.04	4.56	4.70	0.982
0.1	0.9	1	2	5.67 ± 0.06	5.75	5.89	0.883
0.1	0.9	0.99	1	5.51 ± 0.06	5.55	5.61	0.901
0.1	0.9	0.9	2	4.03 ± 0.01	4.10	—	0.883
0.01	0.9	1	3	9.49 ± 0.44	10.79	15.91	0.876
0.01	0.9	0.99	2	8.08 ± 0.28	8.55	8.96	0.880
0.01	0.9	0.9	3	4.66 ± 0.01	4.70	—	0.876

Training details. Each demonstration token holds an input beside its label, and two indicator coordinates mark the token type; there is no positional encoding. The Transformer has 12 layers, 8 heads, and width 256. It is trained with Adam (learning rate 10^{-4}) for 40,000 steps at batch size 1,024, on fresh prompts at every step and without a curriculum. Without a description, the description token is masked out of attention, and the prediction is read at the query token. The labels and the description enter divided by $\sqrt{a_0}$; gradients are clipped to norm 1, and training uses bfloat16 autocast. The number of

demonstrations in a training prompt is uniform on 0 to 20 (0 to 15 for the $r = 0.05$ and $r = 0.5$ networks below). Four further networks, at $r = 0.5$ ($p = 1$ and 0.9) and $r = 0.05$ ($p = 1$ and 0.9), reach an ESS within 0.03 of the Bayes-optimal ESS (2.64 against 2.66, 2.16 against 2.15, 7.32 against 7.35, and 4.38 against 4.37). Without a description, every network's error is within 0.21 of the Bayes-optimal predictor's (about 50) at $n = 0$ and within 0.3 on average over the numbers of demonstrations it was trained on.

ESS as demonstrations accumulate. Figure 5 gives each network's ESS_n at its training reliability. For the loose description it follows the Bayes-optimal ESS_n to within 0.6 at every n the grid can show. For the specific one it falls short by up to 2.4, 2.9, and 1.4 demonstrations at $p = 1$, 0.99, and 0.9, and exceeds it by at most 0.3, until the crossing leaves the grid; with five demonstrations in hand, the three $p = 0.9$ networks have an ESS of 4.83, 4.79, and 4.77, against 5.20.

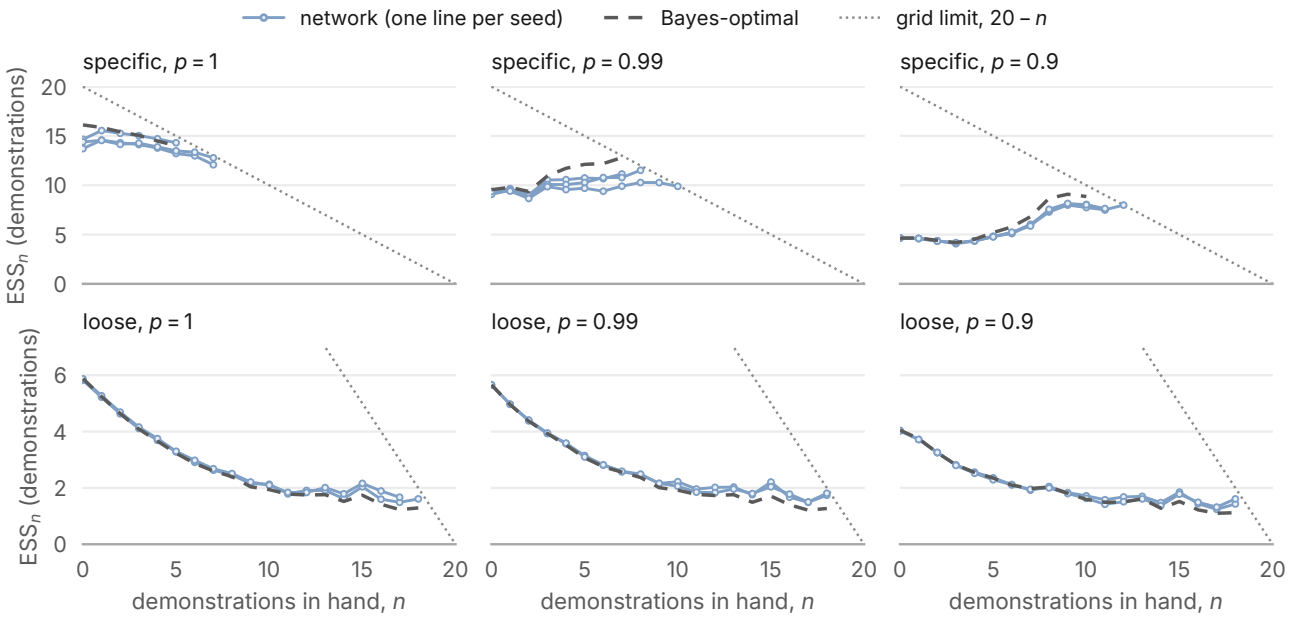


Figure 5. **The networks' ESS as demonstrations accumulate.** ESS_n of each meta-trained network (one line per training seed) and of the Bayes-optimal predictor on the same prompts, trained and tested at the same reliability p, for the specific ($r = 0.01$) and the loose ($r = 0.1$) description; $d = 5$, 20,000 prompts. A curve stops where the crossing lies beyond the 20 demonstrations of the grid; the dotted line, $20 - n$, is the largest ESS the grid can show.

C. The Language-Model Test in Full

Table 3 gives the headline values with their intervals for the six models (gpt-oss-20b under both readouts). Figure 6 shows the error curves the ESS is read from, Tables 11 and 12 the ESS of every description at every number of demonstrations shown, Table 4 the headline ESS under other readings of the curves, and Table 13 the weight on the stated value at every number of demonstrations shown, for the three models read from their distribution.

Models. MiniCPM5 2B (MiniCPM Team, 2026), OLMo 2 7B Instruct, and OLMo 2 13B Instruct begin only 62%, 20%, and 10% of their replies with a number, often explaining instead ("To predict…", "Given…") despite the instruction to reply with the number only, and are not reported beyond that (OLMo 2 7B's $c = 36$ ESS reads 21, from a readout resting on a minority of its probability); we report a model only where it begins at least three quarters of its replies with a number overall, and at least half of its no-description replies over $n = 1$, 2, and 4, the curve its ESS is read from. Thinking is disabled where the chat template allows it, and we start gpt-oss-20b's reply in its final channel. The models run in Hugging Face Transformers at their native precision (`openai/gpt-oss-20b`, `microsoft/phi-4`, `allenai/OLMo-2-0325-32B-Instruct`, `Qwen/Qwen3.5-9B`, `google/gemma-4-12B-it`, `google/gemma-4-31B-it`), and tasks come from a seeded generator.

Prompt. The user turn of the model's chat template:

Table 3. The ESS and the weight on the stated value for each model. Left: the ESS of a description (median over $n = 1$, 2, and 4 demonstrations shown) for a relevant description with $c = 1$, 4, or 36, and for the $c = 36$ description when irrelevant or with a stated 10% or 50% chance of being wrong. Brackets, 95% bootstrap intervals; $\leq$, an upper bound, where one demonstration alone beats the description; $\sigma = 60$, 2,000 tasks; "written": read from the number the model writes. Right: the weight on the stated value with one demonstration shown; the Bayes-optimal weight is $c/(c+n)$. The Bayes-optimal predictor takes a description as relevant unless a chance of being wrong is stated, and then hedges with that chance.

	ESS, median over $n = 1, 2, 4$						weight, $n = 1$		
	relevant			$c = 36$			relevant		
Model	$c = 1$	$c = 4$	$c = 36$	irrelevant	stated 10%	stated 50%	$c = 1$	$c = 4$	$c = 36$
Bayes-optimal	1.3	4.4	36	≤ -1.0	22	5.8	0.50	0.80	0.97
gpt-oss-20b	0.9 [0.7, 1.0]	2.0 [1.7, 2.3]	2.8 [2.6, 3.1]	≤ -0.4	2.4 [1.9, 2.6]	2.3 [1.9, 2.7]	0.53	0.48	0.49
Phi-4	0.7 [0.4, 0.9]	4.1 [3.5, 4.6]	7.6 [6.5, 8.6]	≤ -1.0	8.7 [7.8, 9.8]	9.7 [8.7, 11]	0.82	0.76	0.74
OLMo 2 32B	1.9 [1.7, 2.1]	3.4 [3.2, 3.9]	5.3 [4.5, 6.3]	≤ -0.6	6.0 [5.6, 7.1]	5.4 [5.0, 6.0]	0.72	0.70	0.70
gpt-oss-20b (written)	0.2 [0.1, 0.4]	2.0 [1.9, 2.6]	3.7 [3.0, 4.4]	≤ -1.0	2.4 [2.1, 2.6]	2.3 [2.1, 2.6]	0.64	0.52	0.52
Qwen3.5 9B (written)	0.2 [−0.1, 0.4]	2.7 [2.1, 3.2]	4.0 [3.5, 4.5]	≤ -1.0	0.6 [0.3, 1.0]	1.0 [0.6, 1.5]	0.81	0.76	0.80
Gemma 4 12B (written)	0.0 [−0.1, 0.1]	0.5 [0.3, 0.7]	1.8 [1.4, 2.2]	≤ -0.7	1.3 [1.1, 1.6]	2.7 [2.5, 3.1]	0.34	0.24	0.59
Gemma 4 31B (written)	0.0 [0.0, 0.0]	0.2 [0.1, 0.3]	1.4 [1.2, 1.6]	≤ -0.7	0.6 [0.5, 0.7]	0.5 [0.4, 0.6]	0.48	0.49	0.72

```
The numbers below are drawn from a normal distribution with standard deviation 60.
Its mean was drawn from a normal distribution with mean 510 and standard deviation 30.
548
431
Predict the next number.  Reply with the number only.
```

Without a description, the second sentence is "Its mean is unknown." With a stated reliability, a third sentence follows: "There is a 10% chance that this statement is wrong, in which case the mean is unknown." With no demonstrations, the question is "Predict the first number." The stated value of a relevant description is drawn given the task, so that $m \sim \mathcal{N}(500, 100^2 - \tau^2)$ and $w \mid m \sim \mathcal{N}(m, \tau^2)$; a task is redrawn in the rare case (under 1%) that a number or stated value falls outside 100 to 999.

Measurement. The ESS of §7 is Definition 1 with squared error. Its summary leaves out $n = 0$, where without a description the models rarely answer with a number, and large n, where the ESS mixes the description's value with the models' slow learning (error 9 to 23 after 64 demonstrations at $\sigma = 60$, against 7.5 for the Bayes-optimal predictor) and their over-reliance on loose descriptions. A model that learns slowly from demonstrations values a description more in its own units; the weights do not depend on this. The no-description mean squared error at $n = 1, 2, 4, \ldots, 64$ is interpolated linearly in $\log n$, on the squared error, not its root; ESS_n is the first t at which that curve falls to the description's mean squared error at n, minus n. Since the curve starts at one demonstration, when one demonstration alone beats the description we report the bound $\mathrm{ESS}_n \leq 1 - n$. Each bootstrap replicate resamples tasks, the same tasks for both curves, and repeats the reading, so the intervals reflect sampling over tasks under this procedure, not the interpolation bias quantified next.

The Bayes-optimal reference. The Bayes-optimal predictor is that of §7, with the mean uniform over 100 to 999, so that without a description its prediction is essentially the mean of the demonstrations shown (the range barely truncates it). A relevant description's exact Bayes-optimal ESS is then c, up to that truncation, that is 1, 4, and 36. Computed the same way as the models' ESS, it reads 1.3, 4.4, and 36 at $\sigma = 60$, and 1.2, 4.7, and 43 at $\sigma = 30$: the $\log n$ interpolation biases it upward (by about 0.2, 0.3, and 1.8 for $c = 1$, 4, and 36 with exact errors), and the finite sample of tasks shifts it further, in either direction (for $c = 36$, down to 36 at $\sigma = 60$ and up to 43 at $\sigma = 30$). The tables report the computed value for the Bayes-optimal predictor and the models alike. Under the prior the tasks are drawn from, $\mathcal{N}(500, 100^2)$, a predictor without the description would hold a prior precision of $\sigma^2/100^2 = 0.36$ demonstrations at $\sigma = 60$ (0.09 at $\sigma = 30$), and a relevant description's exact ESS would be $c - 0.36$, that is 0.64, 3.64, and 35.64, the $(1 - r)\,c$ of §4. We keep the predictor that knows only what the prompt states, since the models are told nothing else.

Reading the curves. Definition 1 interpolates between the numbers of demonstrations at which the error is known, and the grid here doubles n, so how the curve is read between grid points matters. On the Bayes-optimal predictor's exact errors, σ^2/n without the description and $\sigma^2/(c+n)$ with it, interpolation in $\log n$ overstates the ESS by 0.2, 0.3, and 1.8

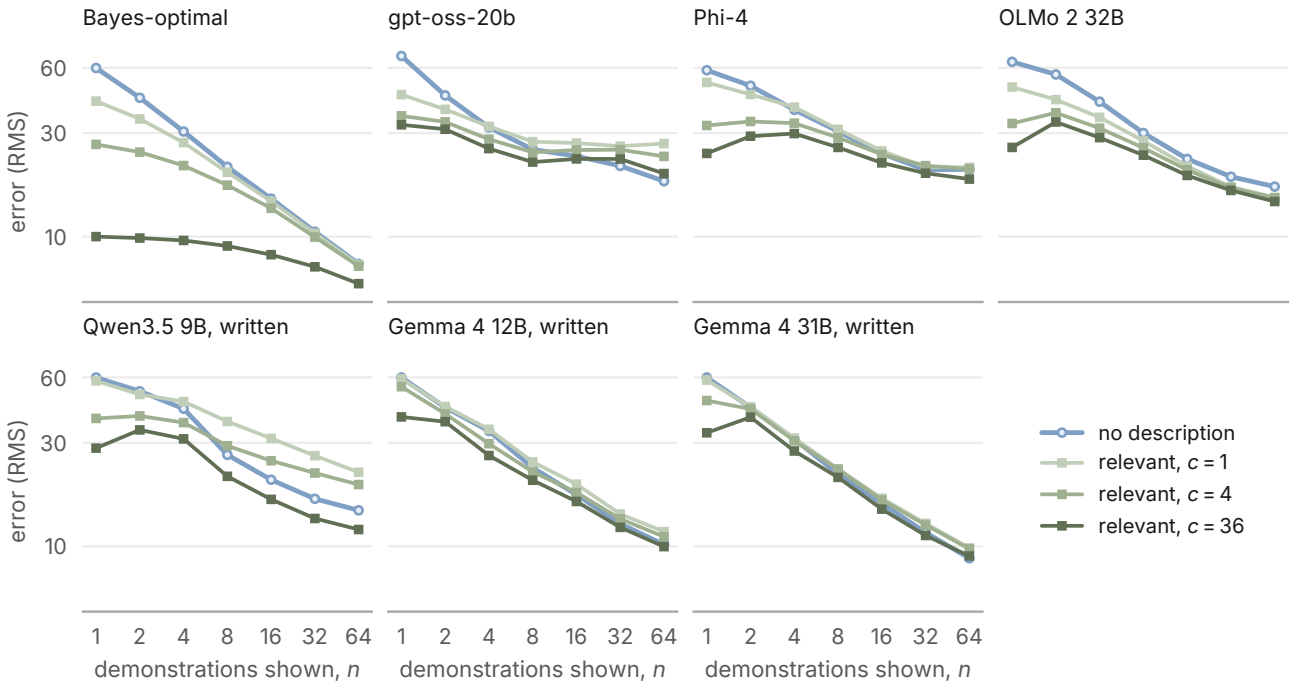


Figure 6. **The error curves the language-model ESS is read from.** Root-mean-square error against the number of demonstrations shown, without a description (blue) and with a relevant description (green, darker for larger c), for the Bayes-optimal predictor and the six reported models; $\sigma = 60$, 2,000 tasks, replies without a number left out. The ESS of a description at n is the horizontal distance from its point at n to the blue curve.

for $c = 1$, 4, and 36, and interpolation in n by 0.3, 0.6, and 4.1. For models answering without reasoning the first crossing is also the only one: of their twelve no-description curves at the two noise levels, only Phi-4's at $\sigma = 60$ rises, from 20.3 at 32 demonstrations to 20.4 at 64. Table 4 gives the headline ESS under four readings: ours; interpolation in n; the curve made non-increasing by a least-squares monotone fit (pool adjacent violators); and the curve $A/(n + \kappa) + B$ fitted to the no-description errors by relative least squares, the form of a Bayesian learner's error plus a floor, which recovers the exact ESS from the Bayes-optimal predictor's exact errors. No model's headline ESS moves by more than 1.4 demonstrations, and none exceeds 8.1 for the $c = 36$ description, against 35 to 38 for the Bayes-optimal predictor. The readings differ more at larger n, where a model's no-description curve flattens and a small difference in error spans many demonstrations: gpt-oss-20b's $c = 36$ ESS at $n = 8$ is 17 in Table 11 and 9.2 from the fitted curve. In Figure 6, the error of four models with the $c = 36$ description rises from one demonstration shown to two, because each demonstration pulls their prediction off the stated value much more than it pulls the Bayes-optimal prediction (Table 13).

Replies. A model's share of replies that begin with a number is the probability its first token puts on the number tokens; where that share is below three quarters, Table 13 sets the cell in italics. The noise-30 conditions, which realize the same three c with both standard deviations in the prompt halved, show the same pattern as the noise-60 ones (Table 11). OLMo 2 32B's first token puts 41% of its probability on numbers in the no-description prompts with one demonstration shown and 75% with two, so its no-description prediction at $n = 1$ rests on a minority of that probability.

Written readout. For Qwen3.5 9B, Gemma 4 12B, and Gemma 4 31B, and as a check on the others, the prediction is the first number with three digits before any decimal part in the model's reply under greedy decoding with eight new tokens, decimals kept, the rule for every text reply here (these runs kept only the integer part, but direct answers rarely have a decimal part: 3 of GPT-5.6-Luna's 1,300 with reasoning off). A reply without one is left out of both the error curves and the weight regression; the share of replies with one is the "number" column. On gpt-oss-20b the written readout gives weights within 0.21 of the distribution readout's in every condition and at every n, and headline ESS medians of 0.2, 2.0, and 3.7 against 0.9, 2.0, and 2.8; it is the noisier readout, since one greedy answer stands in for the distribution's mean. Phi-4 begins only 63% of its written replies with a number over all conditions, so Table 3 reads it from its distribution only; in the wording ablation ($n = 1$, 2, 4) its share is above three quarters for the paper's wording. OLMo 2 32B writes a number

Table 4. **The headline ESS under four readings of the no-description curve.** Median over $n = 1$, 2, and 4 demonstrations shown for the three relevant descriptions; $\sigma = 60$, 2,000 tasks. Log n: the readings of Table 3 (first crossing, interpolation in $\log n$); linear: interpolation in n; monotone: after a monotone fit; fit: the fitted curve $A/(n + \kappa) + B$. The exact Bayes-optimal ESS is 1, 4, and 36.

	$c = 1$				$c = 4$				$c = 36$			
Predictor	log n	linear	monotone	fit	log n	linear	monotone	fit	log n	linear	monotone	fit
Bayes-optimal	1.3	1.4	1.3	1.0	4.4	4.7	4.4	4.0	36	38	36	35
gpt-oss-20b	0.9	1.0	0.9	0.9	2.0	2.4	2.0	2.4	2.8	2.9	2.8	3.4
Phi-4	0.7	0.8	0.7	0.4	4.1	4.4	4.1	3.6	7.6	8.1	7.6	7.1
OLMo 2 32B	1.9	2.1	1.9	1.4	3.4	3.7	3.4	3.3	5.3	5.8	5.3	5.5
Qwen3.5 9B (written)	0.2	0.3	0.2	−0.1	2.7	2.9	2.7	1.3	4.0	4.3	4.0	2.8
Gemma 4 12B (written)	0.0	0.0	0.0	0.0	0.5	0.6	0.5	0.4	1.8	1.9	1.8	1.6
Gemma 4 31B (written)	0.0	0.0	0.0	−0.1	0.2	0.3	0.2	0.0	1.4	1.7	1.4	1.0

Table 5. **ESS under actual unreliability.** The relevant and irrelevant conditions combined with weights p and $1 - p$; median over $n = 1$, 2, and 4 demonstrations shown, with 95% bootstrap intervals over tasks; $\sigma = 60$, 2,000 tasks. A bound where one demonstration alone beats the description at some n.

	irrelevant one time in ten ($p = 0.9$)		irrelevant half the time ($p = 0.5$)	
Predictor	$c = 4$	$c = 36$	$c = 4$	$c = 36$
Bayes-optimal, takes it as relevant	1.0 [0.8, 1.1]	0.2 [0.0, 0.4]	≤ -1.0	≤ -1.0
Bayes-optimal, hedges at p	2.3 [2.1, 2.7]	5.5 [5.1, 6.0]	0.6 [0.5, 0.7]	0.9 [0.8, 1.1]
gpt-oss-20b, not told	1.2 [1.0, 1.4]	1.6 [1.4, 1.7]	0.1 [0.0, 0.2]	≤ 0.0
gpt-oss-20b, told	1.1 [1.0, 1.3]	1.3 [1.1, 1.4]	−0.1 [−0.1, 0.0]	−0.1 [−0.1, 0.0]
Phi-4, not told	1.3 [1.1, 1.5]	1.9 [1.7, 2.4]	≤ -1.0	≤ -1.0
Phi-4, told	2.7 [2.2, 3.2]	3.6 [3.1, 4.1]	≤ 0.0	≤ 0.0
OLMo 2 32B, not told	2.5 [2.2, 2.8]	3.2 [2.9, 3.5]	≤ 0.7	≤ 0.6
OLMo 2 32B, told	3.2 [2.9, 3.4]	4.0 [3.6, 4.3]	≤ 1.0	≤ 0.9
Qwen3.5 9B (written), not told	1.0 [0.7, 1.2]	1.8 [1.3, 2.1]	≤ -0.3	≤ -0.9
Qwen3.5 9B (written), told	−1.0 [−1.0, −0.7]	−0.1 [−0.5, 0.3]	≤ -1.0	≤ -1.0
Gemma 4 12B (written), not told	0.3 [0.2, 0.5]	0.7 [0.5, 1.0]	≤ -0.1	≤ 0.0
Gemma 4 12B (written), told	0.6 [0.5, 0.7]	1.0 [0.9, 1.3]	≤ 0.0	≤ 0.1
Gemma 4 31B (written), not told	0.1 [0.0, 0.2]	0.4 [0.3, 0.5]	≤ 0.0	≤ -0.1
Gemma 4 31B (written), told	≤ 0.0	0.2 [0.1, 0.3]	≤ 0.0	≤ 0.0

in only 5%, 36%, and 48% of its no-description replies at $n = 1$, 2, and 4 ($\sigma = 60$), so its written ESS is not reported; its written weights on the three relevant descriptions at $n = 1$ are 0.83, 0.81, and 0.83, against 0.72, 0.70, and 0.70 from its distribution.

Actual unreliability. The stated-reliability conditions keep the description relevant and test whether a model hedges as told. Table 5 instead realizes the mixture of §3: the squared errors of the relevant and the irrelevant condition, on the same tasks, are combined with weights p and $1 - p$ before the ESS is read, for a model not told the chance and for one told it, against the fully trusting Bayes-optimal predictor and the calibrated one, which hedges at p. With one description in ten irrelevant, the $c = 36$ description is worth 5.5 demonstrations to the calibrated predictor and 0.2 to the fully trusting one; to the models it is worth 0.4 to 3.2 when not told and −0.1 to 4.0 when told. Their flat weight on a description acts as a hedge between full trust and calibration, and being told the chance moves it by at most two demonstrations, in either direction. With half the descriptions irrelevant, the calibrated predictor keeps under one demonstration and no model keeps more than one.

Wording of the description. We reran the three relevant descriptions and the no-description prompt at $n = 1$, 2, and 4 with three other wordings of the description sentence: "Its mean is about m, give or take τ.", "Mean: $m \pm \tau$", and "An earlier estimate put its mean at m, with a standard error of τ." (Table 6; written readout, 2,000 tasks). The wording sets how much a model relies on the description: with one demonstration shown, Gemma 4 12B's weight on the $c = 1$ value is 0.34 under the paper's wording and 1.00 under the other three, and gpt-oss-20b's ranges from 0.17 to 0.67 (a weight above 1 means the prediction moves more than the stated value does). Under no wording does the weight follow the stated specificity

as the Bayes-optimal weight does: the gap between $c = 36$ and $c = 1$ is at most 0.25 at $n = 1$ and 0.44 at $n = 4$ over all models and wordings, against 0.47 and 0.70 for the Bayes-optimal predictor.

Table 6. **The wording of the description sets how much a model relies on it, but under no wording does the weight follow the stated specificity.** Weight on the stated value for the three relevant descriptions at $n = 1$ and 4 under four wordings (written readout, $2{,}000$ tasks). Number: the share of replies with a number; italics: below three quarters in that cell. The Bayes-optimal weight is 0.50, 0.80, and 0.97 at $n = 1$ and 0.20, 0.50, and 0.90 at $n = 4$.

			weight, $n = 1$			weight, $n = 4$		
Model	Wording of the description	number	$c = 1$	$c = 4$	$c = 36$	$c = 1$	$c = 4$	$c = 36$
Gemma 4 12B	normal, mean m, sd τ (the paper's)	1.00	0.34	0.24	0.59	0.19	0.24	0.31
	about m, give or take τ	1.00	1.00	1.00	1.00	0.87	0.76	0.82
	Mean: $m \pm \tau$	1.00	1.00	1.00	1.00	0.60	0.66	0.74
	earlier estimate m, s.e. τ	1.00	1.00	1.00	1.00	0.45	0.54	0.67
Gemma 4 31B	normal, mean m, sd τ (the paper's)	1.00	0.48	0.49	0.72	0.02	0.04	0.18
	about m, give or take τ	1.00	1.00	1.00	1.00	1.01	1.22	1.45
	Mean: $m \pm \tau$	1.00	1.00	1.00	1.00	0.90	1.03	1.15
	earlier estimate m, s.e. τ	1.00	0.94	0.94	0.96	0.09	0.23	0.41
gpt-oss-20b	normal, mean m, sd τ (the paper's)	1.00	0.64	0.52	0.52	0.23	0.29	0.34
	about m, give or take τ	1.00	0.24	0.17	0.38	0.28	0.29	0.38
	Mean: $m \pm \tau$	1.00	0.17	0.26	0.40	0.23	0.24	0.30
	earlier estimate m, s.e. τ	1.00	0.67	0.63	0.67	0.28	0.37	0.42
OLMo 2 32B	normal, mean m, sd τ (the paper's)	0.61	0.83	0.81	0.83	*0.20*	*0.29*	0.38
	about m, give or take τ	0.82	0.93	0.92	0.96	0.60	0.57	0.65
	Mean: $m \pm \tau$	0.82	0.88	0.83	0.88	0.44	0.46	0.55
	earlier estimate m, s.e. τ	0.82	0.95	0.94	0.95	0.49	0.57	0.65
Phi-4	normal, mean m, sd τ (the paper's)	0.77	0.96	0.78	0.76	*0.38*	*0.37*	*0.38*
	about m, give or take τ	0.97	1.02	1.01	1.02	0.65	0.57	0.62
	Mean: $m \pm \tau$	0.83	*0.99*	*0.87*	*0.87*	0.45	0.42	0.50
	earlier estimate m, s.e. τ	0.81	0.88	0.76	0.73	*0.47*	*0.49*	0.54
Qwen3.5 9B	normal, mean m, sd τ (the paper's)	1.00	0.81	0.76	0.80	0.26	0.27	0.44
	about m, give or take τ	1.00	0.99	1.00	0.99	0.35	0.43	0.57
	Mean: $m \pm \tau$	1.00	1.08	1.25	1.14	0.19	0.27	0.38
	earlier estimate m, s.e. τ	1.00	0.87	0.86	0.86	0.61	0.75	0.82

Specificity control. For each task the control states the value drawn for $\tau = 10$, or the one drawn for $\tau = 60$, with each of $\tau = 60$, 30, and 10, on the same $2{,}000$ tasks and demonstrations as the main run, together with its own no-description condition, at every n (Table 7). Its repeated cells reproduce the main run exactly for every model but gpt-oss-20b, whose distribution readout differs by a median of 0.8 to 1.2 (at most 39) and whose written answers differ in 12% of replies (median difference 0), likely from a newer library version. The primary contrast is the paired change in each task's prediction when the stated standard deviation goes from 60 to 10, regressed across tasks and $n = 1$, 2, and 4 on the change in the Bayes-optimal prediction, on the tasks answered in both cells: 1 for a learner that follows the stated specificity as the Bayes-optimal predictor does, 0 for one that ignores it. The Bayes-optimal reference takes the statement at face value, so with the value accurate to $\tau = 60$ but stated as $\tau = 10$ its ESS is about -0.8, and a model that ignores the stated specificity can beat it there. With the stated value fixed, its accuracy, not the stated specificity, mainly sets the ESS of the models answering without reasoning: with the value accurate to $\tau = 60$, each one's ESS is between -0.1 and 2.0 under every stated τ.

Reasoning. Both reasoning runs use the first 200 tasks of the main run and the cells of Table 8: the no-description curve at $n = 1$ to 64 (with $n = 40$ and 48 for Haiku 5.5, where the $c = 36$ ESS is read), and each description and control condition at $n = 1$, 2, and 4. gpt-oss-20b runs in vLLM with greedy decoding at low reasoning effort and up to $4{,}096$ new tokens; 99.9% of its $7{,}400$ replies reach the final channel (at least 97% in every cell), with 322 tokens on average, and the number there, read by the rule of the written readout, is its prediction. An earlier run with 384 tokens finished only 28% to 54% of the replies where a reliability is stated, and the hedge it seemed to show between a 10% and a 50% chance came from which replies finished. Claude Haiku 5.5 (`claude-haiku-5-5`) is called through the Claude Code command line on a

Table 7. **With the stated value and the demonstrations fixed, models answering without reasoning respond to the stated specificity far less than the Bayes-optimal predictor; models that reason follow it.** Specificity control, $\sigma = 60$. Response to the stated sd: the paired change in prediction from a stated standard deviation of 60 to 10, as a share of the Bayes-optimal change, over $n = 1$, 2, and 4, with 95% bootstrap intervals, for a value accurate to $\tau = 10$ or to $\tau = 60$. Right: for the value accurate to $\tau = 10$, the weight on it at $n = 1$ and the ESS (median over $n = 1$, 2, 4) under each stated τ; the Bayes-optimal predictor takes the stated τ at face value. Upper rows: 2,000 tasks, read as in Table 3; lower rows: reasoning, the first 200 tasks, on which the Bayes-optimal ESS is 2.2, 14, and 35.

	response to the stated sd		value accurate to sd 10, stated sd 60 / 30 / 10					
	value sd 10	value sd 60	weight, $n = 1$			ESS		
Bayes-optimal	1	1	0.50	0.80	0.97	2.5	13	36
gpt-oss-20b	0.02 [0.02, 0.03]	0.00 [0.00, 0.01]	0.48	0.47	0.49	2.7	2.6	2.8
Phi-4	−0.13 [−0.14, −0.12]	−0.13 [−0.13, −0.12]	0.81	0.75	0.74	12	7.2	7.6
OLMo 2 32B	0.01 [0.01, 0.01]	0.01 [0.00, 0.01]	0.70	0.70	0.70	4.8	4.6	5.3
gpt-oss-20b (written)	0.01 [−0.01, 0.04]	−0.01 [−0.03, 0.01]	0.52	0.46	0.50	2.9	3.5	3.8
Qwen3.5 9B (written)	0.12 [0.09, 0.14]	0.16 [0.13, 0.18]	0.76	0.75	0.80	3.3	3.5	4.0
Gemma 4 12B (written)	0.19 [0.16, 0.23]	0.12 [0.09, 0.14]	0.38	0.29	0.59	1.0	1.0	1.8
Gemma 4 31B (written)	0.38 [0.34, 0.41]	0.28 [0.25, 0.31]	0.43	0.49	0.72	0.2	0.3	1.4
gpt-oss-20b, low	1.03 [0.93, 1.14]	1.00 [0.96, 1.04]	0.50	0.79	0.95	0.5	8.6	> 60
Haiku 5.5, medium	0.92 [0.84, 0.99]	0.92 [0.86, 0.97]	0.36	0.69	0.74	2.1	11	28
Haiku 5.5, low	0.70 [0.59, 0.79]	0.79 [0.70, 0.86]	0.30	0.34	0.41	2.4	16	26

Table 8. **With reasoning, Haiku 5.5 comes close to the Bayes-optimal ESS; gpt-oss-20b gets the weight right but loses ESS to a few large errors.** First 200 tasks, $\sigma = 60$. Weight on the stated value at $n = 1$; ESS, median over $n = 1$, 2, and 4, with 95% bootstrap intervals, read from the model's own no-description curve, and (ref.) its ESS read from the Bayes-optimal predictor's. $\leq$, an upper bound; > 60, beyond the grid. The Bayes-optimal values are read the same way on the same tasks.

	Bayes-optimal		gpt-oss-20b, low			Haiku 5.5, medium			Haiku 5.5, low		
Description	weight	ESS	weight	ESS	ref.	weight	ESS	ref.	weight	ESS	ref.
$c = 1$	0.50	1.0	0.50	0.3 [−0.2, 1.0]	0.3 [−0.2, 1.0]	0.44	1.1 [0.6, 1.6]	1.1 [0.6, 1.6]	0.36	0.8 [0.5, 1.7]	0.7 [0.4, 1.2]
$c = 4$	0.80	3.5	0.78	1.0 [−0.1, 3.1]	1.0 [−0.1, 3.1]	0.66	3.3 [2.2, 4.7]	3.5 [2.2, 4.8]	0.34	3.4 [2.5, 5.0]	3.3 [2.1, 4.5]
$c = 36$	0.97	35	0.95	> 60 [13, >]	24 [12, 35]	0.74	28 [21, >]	28 [21, 39]	0.41	26 [22, 32]	24 [17, 31]
irrelevant	0.97	≤ -1.0	0.97	≤ -1.0	≤ -1.0	0.87	≤ -1.0	≤ -1.0	0.60	≤ -1.0	≤ -1.0
stated 10%	0.88	27	0.95	1.3 [−0.2, >]	1.2 [−0.3, 15]	0.79	33 [26, >]	33 [26, 40]	0.42	27 [22, 38]	26 [20, 37]
stated 50%	0.58	5.7	0.89	8.8 [4.2, >]	8.8 [4.2, 17]	0.55	3.8 [2.3, 6.4]	3.9 [2.4, 6.5]	0.49	5.7 [4.3, 21]	5.6 [4.2, 9.2]

subscription, one prompt per call, with no tools, no project or user instructions, and a one-line system prompt, at its default (medium) reasoning effort and at low effort; at medium it thinks before most answers (270 thinking tokens on average; 37% to 88% of replies at $n = 1$ over all conditions, and nearly all from $n = 2$), its replies are sampled at the command line's default temperature, and all but one of its 7,800 replies contain a number (three at low effort). Its no-description error is close to the Bayes-optimal predictor's through $n = 48$ (15.8 against 14.8 at $n = 16$, where two replies are far off) and rises at 64, where in 9 of 200 replies it divides the sum by 60 instead of 64. Within each cell, Haiku 5.5's replies split by whether it thinks: at one demonstration, those that think put a weight of 0.50, 0.80, and 0.97 on the relevant $c = 1$, 4, and 36 values, the Bayes-optimal weight to two decimals at either effort, and those that answer directly put 0.00 to 0.05 on it and repeat the number shown. On these three descriptions at low effort it thinks in 34% to 53% of replies, against 72% to 82% at medium, so its overall weight at one demonstration is lower and nearly flat (0.36, 0.34, 0.41); with more demonstrations it thinks more often and its weight is close to the Bayes-optimal again (0.17, 0.48, 0.86 at four, against 0.20, 0.50, 0.90). The choice to think is the model's, so the split describes its replies rather than an experiment. "Ref. ESS" reads the model's error with the description off the Bayes-optimal no-description curve instead of the model's own: the demonstrations a Bayes-optimal learner without the description would need to match it.

A frontier model inside an agent harness. The models above are open ones run locally, so that their first-token distributions can be read. The squared-error ESS needs only answers, so we also put the prompts of three relevant descriptions ($c = 1$, 4, 36; $\sigma = 60$) with $n = 1$ and 4 demonstrations, 100 tasks each, to GPT-5.6-Luna (model `gpt-5.6-luna`) through the Codex command-line agent in its read-only sandbox, on a subscription, once with its reasoning effort set to low (about 115 reasoning tokens per answer) and once with reasoning off. This measures the model as deployed in that agent: its instructions and tool definitions, about 26,000 tokens, surround the prompt, as a provider's system prompt surrounds it through an API. The model used no tool in any of these 1,200 replies, and every reply is a number alone (ten with a trailing

period; 181 with a decimal part, which we keep).

With reasoning, the weight on the stated value is the Bayes-optimal $c/(c+n)$ to within 0.03 in all six cells, the implied worth $n\,a/b$, with a and b the weights on the stated value and on the demonstrations' mean, is c to within half a demonstration (Table 9), and the error matches the Bayes-optimal predictor's to within 0.8. The prompt therefore contains what is needed to reach the Bayes-optimal weight, and this model reaches it when it reasons. With reasoning off, the same model shows the flat weight of §7: 0.72 to 0.88 at $n = 1$ and 0.40 to 0.47 at $n = 4$ whatever the stated specificity, and an implied worth of 2.6 to 5.8 demonstrations for descriptions worth 1 to 36.

To read the ESS of Definition 1 from this model's own no-description curve, we also sent the no-description prompts on the same 100 tasks at $n = 1, 2, 4, \ldots, 64$ in both modes (1,400 replies, all numbers). With reasoning, the model averaged the shown numbers with a shell command in the agent's sandbox (Python in 127 prompts, Node.js or awk in the rest) in 174 of these 700 prompts (4 at $n = 16$, 74 at $n = 32$, and 96 at $n = 64$), and its no-description error matches the Bayes-optimal predictor's within 0.1 at every n (for example 9.4 and 7.0 at $n = 32$ and 64, against 9.4 and 6.9); with reasoning off it used no tool. A more accurate no-description curve lowers the ESS read from it, so the tool use does not inflate the ESS. With reasoning, the ESS of the three descriptions at $n = 1$ and 4 matches the Bayes-optimal predictor's, read the same way from its predictions on the same tasks, to within 0.2 demonstrations for $c = 1$ and 4 and 32 against 33 for $c = 36$ (Table 10; with 100 tasks the reading itself moves the Bayes-optimal 4 to 2.4 and 2.9). With reasoning off, the $c = 36$ description is worth 5.4 demonstrations at $n = 1$ and 1.3 at $n = 4$, in the range of the open models of §7, and the $c = 1$ description hurts at $n = 4$.

Table 9. **A frontier model reaches the Bayes-optimal weight when it reasons and shows the flat weight of the open models when it does not.** GPT-5.6-Luna inside the Codex agent harness, 100 tasks per cell, relevant descriptions at $\sigma = 60$; weight on the stated value and implied worth $n\,a/b$ from the regression of §7; bootstrap standard errors of the weights are at most 0.03 with reasoning and 0.07 without.

	weight on the stated value						implied worth, demonstrations					
	$n = 1$			$n = 4$			$n = 1$			$n = 4$		
Bayes-optimal ESS c	1	4	36	1	4	36	1	4	36	1	4	36
Bayes-optimal	0.50	0.80	0.97	0.20	0.50	0.90	1	4	36	1	4	36
GPT-5.6-Luna, low reasoning	0.50	0.77	0.97	0.21	0.50	0.90	1.0	3.6	35.7	1.0	4.0	36.5
GPT-5.6-Luna, reasoning off	0.79	0.72	0.88	0.43	0.40	0.47	3.6	2.6	5.8	3.1	2.7	3.7

Table 10. **When it reasons, GPT-5.6-Luna's ESS is consistent with the Bayes-optimal ESS.** ESS at $n = 1$ and 4 demonstrations shown, read from the model's own no-description curve ($n = 1$ to 64, the same 100 tasks; with reasoning, the model averaged the numbers with a shell command in 174 of these 700 prompts, mostly at $n = 32$ and 64) as in §7, with 95% bootstrap intervals over tasks; the Bayes-optimal predictor's predictions on the same tasks are read the same way.

	$n = 1$			$n = 4$		
Bayes-optimal ESS c	1	4	36	1	4	36
Bayes-optimal, read the same way	1.0 [0.6, 1.8]	2.4 [1.6, 3.9]	33 [25, 49]	1.7 [0.5, 3.6]	2.9 [1.2, 7.7]	33 [23, 50]
GPT-5.6-Luna, low reasoning	1.1 [0.6, 1.8]	2.3 [1.4, 3.8]	32 [25, 48]	1.5 [0.3, 3.4]	3.0 [1.2, 7.7]	32 [23, 49]
GPT-5.6-Luna, reasoning off	0.1 [0.0, 0.5]	1.2 [0.8, 2.1]	5.4 [2.5, 14]	−2.2 [−2.6, −1.4]	0.1 [−0.7, 2.5]	1.3 [−0.2, 3.8]

Table 11. ESS of a description for every condition and number of demonstrations shown, for the Bayes-optimal predictor and the three models read from their distribution (2,000 tasks). "number" is the share of a model's replies that begin with a three-digit number, averaged over n; "10% wrong" and "50% wrong" mark a stated chance of being wrong. The Bayes-optimal predictor is that of §7, optimal for what the prompt states: it takes an irrelevant description with no stated chance as relevant, so its ESS can be negative. $\leq$, an upper bound; $> k$, the error with the description is below the no-description error at every n measured, up to 64, so the ESS exceeds $k = 64 - n$.

Predictor	Description	number	$n = 1$	$n = 2$	$n = 4$	$n = 8$	$n = 16$	$n = 32$	$n = 64$
Bayes-optimal	$\sigma = 60$, $c = 1$, relevant		1.2	1.3	1.3	1.3	1.2	1.5	> 0
	$\sigma = 60$, $c = 1$, irrelevant		≤ 0.0	-0.5	-0.8	-1.2	-1.7	-1.5	-1.7
	$\sigma = 60$, $c = 4$, relevant		4.5	4.4	3.9	4.6	4.6	5.2	> 0
	$\sigma = 60$, $c = 4$, irrelevant		≤ 0.0	≤ -1.0	≤ -3.0	-6.2	-12	-20	-32
	$\sigma = 60$, $c = 36$, relevant		36	36	37	38	39	> 32	> 0
	$\sigma = 60$, $c = 36$, irrelevant		≤ 0.0	≤ -1.0	≤ -3.0	≤ -7.0	≤ -15	≤ -31	-62
	$\sigma = 30$, $c = 1$, relevant		1.0	1.2	1.4	1.6	1.4	1.5	> 0
	$\sigma = 30$, $c = 4$, relevant		4.5	4.7	5.0	5.1	4.9	5.6	> 0
	$\sigma = 30$, $c = 36$, relevant		42	43	43	43	42	> 32	> 0
	$\sigma = 60$, $c = 4$, relevant, 10% wrong		3.8	3.9	3.6	4.4	4.5	5.1	> 0
	$\sigma = 60$, $c = 4$, irrelevant, 10% wrong		≤ 0.0	-0.7	-1.2	-1.6	-2.2	-1.9	-2.2
	$\sigma = 60$, $c = 36$, relevant, 10% wrong		13	22	26	35	38	> 32	> 0
	$\sigma = 60$, $c = 36$, irrelevant, 10% wrong		≤ 0.0	-0.9	-1.7	-2.8	-5.4	-6.9	-10
	$\sigma = 60$, $c = 4$, relevant, 50% wrong		1.9	2.2	2.9	3.7	4.1	4.5	> 0
	$\sigma = 60$, $c = 4$, irrelevant, 50% wrong		≤ 0.0	-0.2	-0.4	-0.5	-0.8	-0.6	-0.7
	$\sigma = 60$, $c = 36$, relevant, 50% wrong		3.1	5.8	12	21	31	> 32	> 0
	$\sigma = 60$, $c = 36$, irrelevant, 50% wrong		≤ 0.0	-0.3	-0.7	-1.2	-2.4	-2.8	-4.5
gpt-oss-20b	$\sigma = 60$, $c = 1$, relevant	0.98	1.0	0.9	-0.1	-1.5	-9.2	-25	-57
	$\sigma = 60$, $c = 1$, irrelevant	0.98	≤ 0.0	-0.3	-1.5	-3.6	-10	-25	-59
	$\sigma = 60$, $c = 4$, relevant	0.98	2.3	1.7	2.0	2.9	-7.5	-24	-48
	$\sigma = 60$, $c = 4$, irrelevant	0.97	≤ 0.0	-0.3	-1.4	-3.7	-11	-26	-59
	$\sigma = 60$, $c = 36$, relevant	0.98	2.8	2.3	3.8	17	3.7	-12	-17
	$\sigma = 60$, $c = 36$, irrelevant	0.98	≤ 0.0	-0.5	-1.5	-3.7	-11	-26	-60
	$\sigma = 30$, $c = 1$, relevant	0.99	0.8	0.4	-0.3	-1.8	-9.9	-26	-59
	$\sigma = 30$, $c = 4$, relevant	0.98	1.2	1.3	1.9	3.4	-8.1	-24	-44
	$\sigma = 30$, $c = 36$, relevant	0.99	1.6	2.3	4.7	31	3.0	-13	> 0
	$\sigma = 60$, $c = 4$, relevant, 10% wrong	0.96	2.2	1.8	0.0	-0.2	-8.1	-24	-52
	$\sigma = 60$, $c = 4$, irrelevant, 10% wrong	0.96	≤ 0.0	-0.6	-1.1	-2.2	-9.6	-25	-58
	$\sigma = 60$, $c = 36$, relevant, 10% wrong	0.97	2.4	2.4	0.5	2.8	-3.6	-20	-33
	$\sigma = 60$, $c = 36$, irrelevant, 10% wrong	0.96	≤ 0.0	-0.7	-1.1	-2.3	-9.7	-25	-59
	$\sigma = 60$, $c = 4$, relevant, 50% wrong	0.95	2.3	1.7	0.1	-0.5	-8.3	-24	-53
	$\sigma = 60$, $c = 4$, irrelevant, 50% wrong	0.94	≤ 0.0	-0.5	-1.1	-2.3	-9.7	-25	-58
	$\sigma = 60$, $c = 36$, relevant, 50% wrong	0.95	2.8	2.3	0.7	1.2	-4.5	-21	-37
	$\sigma = 60$, $c = 36$, irrelevant, 50% wrong	0.95	≤ 0.0	-0.6	-1.1	-2.3	-9.7	-25	-59
Phi-4	$\sigma = 60$, $c = 1$, relevant	0.87	0.8	0.7	-0.2	-0.4	-1.0	-3.1	-35
	$\sigma = 60$, $c = 1$, irrelevant	0.86	≤ 0.0	≤ -1.0	≤ -3.0	-4.9	-6.1	-8.8	-40
	$\sigma = 60$, $c = 4$, relevant	0.85	5.8	4.1	2.4	2.1	0.7	-4.6	-33
	$\sigma = 60$, $c = 4$, irrelevant	0.84	≤ 0.0	≤ -1.0	≤ -3.0	-4.7	-6.2	-13	-42
	$\sigma = 60$, $c = 36$, relevant	0.86	15	7.6	4.7	5.6	8.5	> 32	> 0
	$\sigma = 60$, $c = 36$, irrelevant	0.84	≤ 0.0	≤ -1.0	≤ -3.0	-4.5	-6.4	-12	-42
	$\sigma = 30$, $c = 1$, relevant	0.87	1.1	0.8	-0.3	-0.3	-1.0	-1.8	-15
	$\sigma = 30$, $c = 4$, relevant	0.86	4.8	3.5	2.0	1.6	0.6	-1.9	-2.2
	$\sigma = 30$, $c = 36$, relevant	0.88	9.7	6.0	3.8	4.4	8.4	17	> 0
	$\sigma = 60$, $c = 4$, relevant, 10% wrong	0.87	6.0	5.5	5.3	8.3	15	> 32	> 0
	$\sigma = 60$, $c = 4$, irrelevant, 10% wrong	0.85	≤ 0.0	≤ -1.0	-2.1	-1.2	2.9	-3.2	> 0
	$\sigma = 60$, $c = 36$, relevant, 10% wrong	0.87	10	8.7	8.2	17	> 48	> 32	> 0
	$\sigma = 60$, $c = 36$, irrelevant, 10% wrong	0.86	≤ 0.0	≤ -1.0	-2.4	-1.7	1.3	-4.7	> 0
	$\sigma = 60$, $c = 4$, relevant, 50% wrong	0.87	6.0	6.4	7.7	17	> 48	> 32	> 0
	$\sigma = 60$, $c = 4$, irrelevant, 50% wrong	0.85	≤ 0.0	≤ -1.0	-1.3	1.5	16	> 32	> 0
	$\sigma = 60$, $c = 36$, relevant, 50% wrong	0.88	9.2	9.7	11	> 56	> 48	> 32	> 0
	$\sigma = 60$, $c = 36$, irrelevant, 50% wrong	0.86	≤ 0.0	≤ -1.0	-1.8	0.6	13	> 32	> 0
OLMo 2 32B	$\sigma = 60$, $c = 1$, relevant	0.90	1.9	1.8	2.0	2.3	5.8	> 32	> 0
	$\sigma = 60$, $c = 1$, irrelevant	0.89	≤ 0.0	-0.5	-0.2	-0.2	1.7	25	> 0
	$\sigma = 60$, $c = 4$, relevant	0.91	5.8	3.4	3.4	4.5	9.3	> 32	> 0
	$\sigma = 60$, $c = 4$, irrelevant	0.90	≤ 0.0	-0.7	-0.3	-0.5	0.3	17	> 0
	$\sigma = 60$, $c = 36$, relevant	0.91	11	4.6	5.3	6.8	15	> 32	> 0
	$\sigma = 60$, $c = 36$, irrelevant	0.90	≤ 0.0	≤ -1.0	-0.6	-0.5	-0.2	15	> 0
	$\sigma = 30$, $c = 1$, relevant	0.92	1.8	1.9	1.7	1.7	4.4	24	> 0
	$\sigma = 30$, $c = 4$, relevant	0.92	4.8	3.3	3.1	4.0	9.5	> 32	> 0
	$\sigma = 30$, $c = 36$, relevant	0.93	7.0	4.4	4.0	5.9	14	> 32	> 0
	$\sigma = 60$, $c = 4$, relevant, 10% wrong	0.96	6.0	4.6	3.8	4.5	5.8	23	> 0
	$\sigma = 60$, $c = 4$, irrelevant, 10% wrong	0.96	≤ 0.0	-0.9	0.0	0.0	-0.4	4.1	> 0
	$\sigma = 60$, $c = 36$, relevant, 10% wrong	0.96	12	5.9	6.0	6.3	8.9	31	> 0
	$\sigma = 60$, $c = 36$, irrelevant, 10% wrong	0.96	≤ 0.0	≤ -1.0	-0.3	-0.1	-0.8	0.3	> 0
	$\sigma = 60$, $c = 4$, relevant, 50% wrong	0.93	5.3	4.2	3.5	4.0	5.4	27	> 0
	$\sigma = 60$, $c = 4$, irrelevant, 50% wrong	0.92	≤ 0.0	-0.5	0.1	0.1	0.0	11	> 0
	$\sigma = 60$, $c = 36$, relevant, 50% wrong	0.93	7.8	5.4	4.9	5.6	7.9	> 32	> 0
	$\sigma = 60$, $c = 36$, irrelevant, 50% wrong	0.92	≤ 0.0	-0.8	-0.1	0.0	-0.3	7.8	> 0

Table 12. ESS of a description for every condition and number of demonstrations shown, for the three models read from the number they write (2,000 tasks; a reply without a number is left out); columns and the Bayes-optimal values as in Table 11.

Predictor	Description	number	$n=1$	$n=2$	$n=4$	$n=8$	$n=16$	$n=32$	$n=64$
Qwen3.5 9B (written)	$\sigma=60, c=1$, relevant	1.00	0.2	0.3	-0.9	-2.8	-9.3	-24	-51
	$\sigma=60, c=1$, irrelevant	1.00	≤ 0.0	≤ -1.0	-2.7	-4.8	-9.7	-23	-49
	$\sigma=60, c=4$, relevant	1.00	3.9	2.7	1.3	-0.7	-6.2	-18	-44
	$\sigma=60, c=4$, irrelevant	1.00	≤ 0.0	≤ -1.0	-1.7	-3.8	-9.4	-24	-49
	$\sigma=60, c=36$, relevant	1.00	6.5	4.0	2.8	6.9	17	>32	>0
	$\sigma=60, c=36$, irrelevant	1.00	≤ 0.0	≤ -1.0	-2.8	-5.1	-10	-24	-51
	$\sigma=30, c=1$, relevant	1.00	≤ 0.0	0.2	-0.7	-2.5	-8.7	-22	-49
	$\sigma=30, c=4$, relevant	1.00	3.4	2.7	1.5	-0.3	-4.5	-14	-40
	$\sigma=30, c=36$, relevant	1.00	4.7	4.3	3.0	9.2	14	>32	>0
	$\sigma=60, c=4$, relevant, 10% wrong	1.00	3.8	-0.4	-0.7	-1.8	-8.5	-21	-47
	$\sigma=60, c=4$, irrelevant, 10% wrong	1.00	≤ 0.0	≤ -1.0	≤ -3.0	-4.9	-10	-24	-50
	$\sigma=60, c=36$, relevant, 10% wrong	1.00	5.3	0.4	0.6	-0.3	-3.3	-4.2	-6.9
	$\sigma=60, c=36$, irrelevant, 10% wrong	1.00	≤ 0.0	≤ -1.0	≤ -3.0	-5.2	-10	-25	-53
	$\sigma=60, c=4$, relevant, 50% wrong	1.00	4.5	-0.1	-0.4	-1.9	-9.0	-23	-50
	$\sigma=60, c=4$, irrelevant, 50% wrong	1.00	≤ 0.0	≤ -1.0	≤ -3.0	-6.2	-12	-26	-54
	$\sigma=60, c=36$, relevant, 50% wrong	1.00	7.3	0.8	1.0	-0.1	-4.0	-4.2	-12
	$\sigma=60, c=36$, irrelevant, 50% wrong	1.00	≤ 0.0	≤ -1.0	≤ -3.0	-7.0	-13	-27	-56
Gemma 4 12B (written)	$\sigma=60, c=1$, relevant	1.00	0.0	0.0	-0.1	-0.6	-3.2	-5.4	-21
	$\sigma=60, c=1$, irrelevant	1.00	≤ 0.0	-0.4	-1.3	-1.3	-4.4	-6.8	-24
	$\sigma=60, c=4$, relevant	1.00	0.3	0.5	1.4	1.1	-0.6	-2.7	-13
	$\sigma=60, c=4$, irrelevant	1.00	≤ 0.0	-0.4	-1.1	-1.2	-4.1	-6.9	-23
	$\sigma=60, c=36$, relevant	1.00	1.8	1.2	2.7	3.6	3.6	4.9	>0
	$\sigma=60, c=36$, irrelevant	1.00	≤ 0.0	-0.8	-1.3	-1.3	-4.9	-7.9	-26
	$\sigma=30, c=1$, relevant	1.00	≤ 0.0	-0.2	-0.1	-0.5	-3.2	-7.9	-27
	$\sigma=30, c=4$, relevant	1.00	0.7	0.8	2.1	1.5	-0.2	-4.6	-18
	$\sigma=30, c=36$, relevant	1.00	4.0	1.7	3.4	3.7	2.4	-1.8	-8.9
	$\sigma=60, c=4$, relevant, 10% wrong	1.00	0.1	0.7	1.4	0.4	-1.0	-3.1	-7.3
	$\sigma=60, c=4$, irrelevant, 10% wrong	1.00	≤ 0.0	-0.2	-0.4	-0.5	-2.2	-5.4	-13
	$\sigma=60, c=36$, relevant, 10% wrong	1.00	0.6	1.3	2.3	1.6	0.7	-0.6	-1.3
	$\sigma=60, c=36$, irrelevant, 10% wrong	1.00	≤ 0.0	-0.2	-0.5	-0.5	-2.5	-5.8	-15
	$\sigma=60, c=4$, relevant, 50% wrong	1.00	1.0	0.6	1.7	1.0	-0.1	-1.7	-4.5
	$\sigma=60, c=4$, irrelevant, 50% wrong	1.00	≤ 0.0	-0.3	-0.4	-0.4	-1.3	-4.1	-9.6
	$\sigma=60, c=36$, relevant, 50% wrong	1.00	5.1	1.0	2.7	2.3	2.4	1.1	>0
	$\sigma=60, c=36$, irrelevant, 50% wrong	1.00	≤ 0.0	-0.3	-0.6	-0.4	-1.9	-4.8	-13
Gemma 4 31B (written)	$\sigma=60, c=1$, relevant	1.00	0.1	0.0	-0.1	-0.5	-1.4	-5.1	-13
	$\sigma=60, c=1$, irrelevant	1.00	≤ 0.0	-0.1	-0.1	-0.5	-1.0	-4.5	-11
	$\sigma=60, c=4$, relevant	1.00	0.8	0.1	0.2	-0.5	-1.0	-4.3	-12
	$\sigma=60, c=4$, irrelevant	1.00	≤ 0.0	-0.2	-0.2	-0.4	-1.0	-4.4	-10
	$\sigma=60, c=36$, relevant	1.00	2.6	0.6	1.4	0.9	2.9	3.4	-2.9
	$\sigma=60, c=36$, irrelevant	1.00	≤ 0.0	≤ -1.0	-0.8	-0.6	-1.1	-2.7	-4.8
	$\sigma=30, c=1$, relevant	1.00	≤ 0.0	0.0	-0.1	-0.7	-1.2	-3.0	-8.1
	$\sigma=30, c=4$, relevant	1.00	0.5	0.0	0.1	-0.5	-1.3	-3.4	-8.3
	$\sigma=30, c=36$, relevant	1.00	4.3	1.5	2.1	2.3	3.3	3.3	-0.4
	$\sigma=60, c=4$, relevant, 10% wrong	1.00	0.1	0.0	0.0	-0.7	-1.9	-5.0	-15
	$\sigma=60, c=4$, irrelevant, 10% wrong	1.00	≤ 0.0	0.0	-0.1	-0.4	-1.2	-4.1	-9.5
	$\sigma=60, c=36$, relevant, 10% wrong	1.00	1.0	0.0	0.6	-0.3	-1.4	-4.8	-12
	$\sigma=60, c=36$, irrelevant, 10% wrong	1.00	≤ 0.0	0.0	-0.2	-0.4	-0.9	-3.0	-6.2
	$\sigma=60, c=4$, relevant, 50% wrong	1.00	0.2	0.0	0.0	-0.7	-1.7	-4.6	-17
	$\sigma=60, c=4$, irrelevant, 50% wrong	1.00	≤ 0.0	0.0	-0.2	-0.4	-1.2	-3.6	-12
	$\sigma=60, c=36$, relevant, 50% wrong	1.00	1.0	0.0	0.5	-0.3	-1.5	-4.5	-15
	$\sigma=60, c=36$, irrelevant, 50% wrong	1.00	≤ 0.0	0.0	-0.1	-0.4	-1.0	-2.9	-8.5

Table 13. Weight on the stated value for every condition and number of demonstrations shown, for the Bayes-optimal predictor and the three models read from their distribution; "number" as in Table 11. A cell is in italics where the model begins fewer than three quarters of its replies with a number at that n.

Predictor	Description	number	$n=0$	$n=1$	$n=2$	$n=4$	$n=8$	$n=16$	$n=32$	$n=64$
Bayes-optimal	$\sigma=60, c=1$, relevant		1.00	0.50	0.33	0.20	0.11	0.06	0.03	0.02
	$\sigma=60, c=1$, irrelevant		1.00	0.50	0.33	0.20	0.11	0.06	0.03	0.02
	$\sigma=60, c=4$, relevant		1.00	0.80	0.67	0.50	0.33	0.20	0.11	0.06
	$\sigma=60, c=4$, irrelevant		1.00	0.80	0.67	0.50	0.33	0.20	0.11	0.06
	$\sigma=60, c=36$, relevant		1.00	0.97	0.95	0.90	0.82	0.69	0.53	0.36
	$\sigma=60, c=36$, irrelevant		1.00	0.97	0.95	0.90	0.82	0.69	0.53	0.36
	$\sigma=30, c=1$, relevant		1.00	0.50	0.33	0.20	0.11	0.06	0.03	0.02
	$\sigma=30, c=4$, relevant		1.00	0.80	0.67	0.50	0.33	0.20	0.11	0.06
	$\sigma=30, c=36$, relevant		1.00	0.97	0.95	0.90	0.82	0.69	0.53	0.36
	$\sigma=60, c=4$, relevant, 10% wrong		0.90	0.71	0.59	0.45	0.30	0.19	0.10	0.05
	$\sigma=60, c=4$, irrelevant, 10% wrong		0.90	0.27	0.14	0.08	0.04	0.02	0.01	0.00
	$\sigma=60, c=36$, relevant, 10% wrong		0.90	0.84	0.84	0.82	0.77	0.65	0.51	0.34
	$\sigma=60, c=36$, irrelevant, 10% wrong		0.90	0.26	0.14	0.07	0.03	0.01	0.00	0.00
	$\sigma=60, c=4$, relevant, 50% wrong		0.50	0.46	0.39	0.32	0.22	0.14	0.08	0.04
	$\sigma=60, c=4$, irrelevant, 50% wrong		0.50	0.12	0.07	0.03	0.02	0.01	0.00	0.00
	$\sigma=60, c=36$, relevant, 50% wrong		0.50	0.55	0.58	0.60	0.59	0.53	0.42	0.29
	$\sigma=60, c=36$, irrelevant, 50% wrong		0.50	0.12	0.06	0.03	0.01	0.00	0.00	0.00
gpt-oss-20b	$\sigma=60, c=1$, relevant	0.98	0.94	0.53	0.29	0.23	0.16	0.12	0.12	0.23
	$\sigma=60, c=1$, irrelevant	0.98	0.93	0.59	0.25	0.18	0.09	0.05	0.04	0.10
	$\sigma=60, c=4$, relevant	0.98	0.92	0.48	0.32	0.26	0.21	0.19	0.18	0.31
	$\sigma=60, c=4$, irrelevant	0.97	0.92	0.57	0.25	0.16	0.08	0.04	0.04	0.09
	$\sigma=60, c=36$, relevant	0.98	0.94	0.49	0.35	0.30	0.25	0.23	0.22	0.35
	$\sigma=60, c=36$, irrelevant	0.98	0.95	0.60	0.25	0.15	0.07	0.04	0.04	0.09
	$\sigma=30, c=1$, relevant	0.99	0.98	0.43	0.23	0.17	0.12	0.09	0.11	0.21
	$\sigma=30, c=4$, relevant	0.98	0.98	0.38	0.28	0.23	0.19	0.18	0.19	0.31
	$\sigma=30, c=36$, relevant	0.99	0.99	0.37	0.32	0.27	0.23	0.22	0.24	0.37
	$\sigma=60, c=4$, relevant, 10% wrong	0.96	0.91	0.45	0.33	0.15	0.13	0.12	0.13	0.22
	$\sigma=60, c=4$, irrelevant, 10% wrong	0.96	0.91	0.55	0.34	0.11	0.05	0.02	0.02	0.04
	$\sigma=60, c=36$, relevant, 10% wrong	0.97	0.92	0.44	0.34	0.17	0.15	0.15	0.16	0.25
	$\sigma=60, c=36$, irrelevant, 10% wrong	0.96	0.92	0.53	0.34	0.10	0.04	0.02	0.02	0.04
	$\sigma=60, c=4$, relevant, 50% wrong	0.95	0.79	0.46	0.32	0.16	0.12	0.12	0.12	0.20
	$\sigma=60, c=4$, irrelevant, 50% wrong	0.94	0.79	0.53	0.32	0.11	0.04	0.02	0.01	0.04
	$\sigma=60, c=36$, relevant, 50% wrong	0.95	0.83	0.46	0.32	0.17	0.14	0.15	0.15	0.23
	$\sigma=60, c=36$, irrelevant, 50% wrong	0.95	0.83	0.53	0.31	0.11	0.04	0.02	0.02	0.04
Phi-4	$\sigma=60, c=1$, relevant	0.87	*0.97*	0.82	0.62	0.43	0.28	0.18	0.13	0.12
	$\sigma=60, c=1$, irrelevant	0.86	*0.97*	0.83	0.64	0.42	0.22	0.11	0.06	0.05
	$\sigma=60, c=4$, relevant	0.85	*0.98*	0.76	0.55	0.37	0.26	0.19	0.15	0.16
	$\sigma=60, c=4$, irrelevant	0.84	*0.98*	0.79	0.56	0.36	0.18	0.09	0.06	0.05
	$\sigma=60, c=36$, relevant	0.86	*0.99*	0.74	0.53	0.38	0.29	0.23	0.20	0.20
	$\sigma=60, c=36$, irrelevant	0.84	*0.99*	0.78	0.53	0.33	0.16	0.09	0.06	0.04
	$\sigma=30, c=1$, relevant	0.87	*0.99*	0.77	0.58	0.38	0.22	0.13	0.09	0.09
	$\sigma=30, c=4$, relevant	0.86	*1.00*	0.69	0.51	0.34	0.23	0.16	0.13	0.13
	$\sigma=30, c=36$, relevant	0.88	*1.00*	0.65	0.48	0.32	0.23	0.18	0.17	0.16
	$\sigma=60, c=4$, relevant, 10% wrong	0.87	*0.94*	0.62	0.46	0.30	0.21	0.15	0.13	0.11
	$\sigma=60, c=4$, irrelevant, 10% wrong	0.85	*0.95*	0.62	0.44	0.26	0.13	0.06	0.04	0.03
	$\sigma=60, c=36$, relevant, 10% wrong	0.87	*0.95*	0.61	0.46	0.33	0.25	0.20	0.18	0.16
	$\sigma=60, c=36$, irrelevant, 10% wrong	0.86	*0.96*	0.61	0.43	0.26	0.13	0.06	0.04	0.03
	$\sigma=60, c=4$, relevant, 50% wrong	0.87	*0.92*	0.57	0.43	0.29	0.20	0.15	0.13	0.11
	$\sigma=60, c=4$, irrelevant, 50% wrong	0.85	*0.92*	0.57	0.42	0.24	0.12	0.06	0.04	0.03
	$\sigma=60, c=36$, relevant, 50% wrong	0.88	*0.93*	0.56	0.44	0.33	0.24	0.21	0.19	0.17
	$\sigma=60, c=36$, irrelevant, 50% wrong	0.86	*0.94*	0.57	0.41	0.24	0.12	0.06	0.04	0.03
OLMo 2 32B	$\sigma=60, c=1$, relevant	0.90	*0.99*	0.72	0.33	0.25	0.16	0.09	0.05	0.05
	$\sigma=60, c=1$, irrelevant	0.89	*0.99*	0.77	0.29	0.16	0.09	0.05	0.02	0.02
	$\sigma=60, c=4$, relevant	0.91	*1.00*	0.70	0.37	0.30	0.21	0.14	0.08	0.07
	$\sigma=60, c=4$, irrelevant	0.90	*1.00*	0.75	0.29	0.15	0.08	0.04	0.02	0.01
	$\sigma=60, c=36$, relevant	0.91	*1.00*	0.70	0.40	0.34	0.26	0.19	0.12	0.10
	$\sigma=60, c=36$, irrelevant	0.90	*1.00*	0.76	0.29	0.17	0.08	0.04	0.02	0.02
	$\sigma=30, c=1$, relevant	0.92	*0.99*	0.69	0.32	0.23	0.14	0.09	0.04	0.04
	$\sigma=30, c=4$, relevant	0.92	*1.00*	0.65	0.34	0.27	0.19	0.13	0.06	0.06
	$\sigma=30, c=36$, relevant	0.93	*1.00*	0.64	0.36	0.32	0.23	0.16	0.08	0.09
	$\sigma=60, c=4$, relevant, 10% wrong	0.96	1.01	0.72	0.42	0.29	0.18	0.11	0.09	0.05
	$\sigma=60, c=4$, irrelevant, 10% wrong	0.96	1.01	0.81	0.32	0.14	0.06	0.03	0.02	0.01
	$\sigma=60, c=36$, relevant, 10% wrong	0.96	1.01	0.71	0.44	0.33	0.23	0.15	0.13	0.07
	$\sigma=60, c=36$, irrelevant, 10% wrong	0.96	1.01	0.82	0.33	0.15	0.07	0.03	0.02	0.01
	$\sigma=60, c=4$, relevant, 50% wrong	0.93	1.02	0.65	0.39	0.27	0.16	0.09	0.07	0.04
	$\sigma=60, c=4$, irrelevant, 50% wrong	0.92	1.01	0.71	0.29	0.13	0.06	0.02	0.01	0.01
	$\sigma=60, c=36$, relevant, 50% wrong	0.93	1.02	0.64	0.42	0.31	0.21	0.13	0.11	0.06
	$\sigma=60, c=36$, irrelevant, 50% wrong	0.92	1.02	0.72	0.30	0.14	0.06	0.03	0.02	0.01